Geometry Seminar Notes

Topology, Morse Theory, and Riemannian Geometry

Notes taken by Shengxuan Zhang

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Introduction

These are the notes for the "Geometry Seminar" taught by Guchuan Li at Peking University in the spring semester of 2026. The lectures were presented in rotation by four students: Shengxuan Zhang (Lectures 2, 4, 8, 12, and 16), Zhaorui Zhang (Lectures 5, 9, and 13), Fangzhou Luo (Lectures 3, 7, 11, and 15), and Jinchen Zhao (Lectures 1, 6, 10, and 14).

The first part of the semester covered several standalone topics. The second half focused primarily on the key material in Milnor's *Morse Theory*.

Contents

1

Triangulation of Surfaces

2026/3/10

Discuss the triangulation of surfaces.

Step I: Jordan Curve Theorem

Def: A Jordan Curve C is an injective map $\mathbb{S}^1 \to \mathbb{S}^2$ or $\mathbb{S}^1 \to \mathbb{R}^2$. (closed not self-intersecting map).
Jordan Curve Theorem (1887): $\mathbb{S}^2 - C$ has two connected components D_+, D_- and $\partial D_+ = \partial D_- = C$. (This holds for any dim.)

Proof by Algebraic Topology Recall some facts from algebraic topology:

- $H_i(\mathbb{S}^n) = \mathbb{Z}$ if $i = 0, n$, and $H_i(\mathbb{S}^n) = 0$ otherwise;
- By Poincaré duality, $H^i(\mathbb{S}^n) = \mathbb{Z}$ if $i = 0, n$ and $H^i(\mathbb{S}^n) = 0$ otherwise;
- $\tilde{H}_0(\mathbb{S}^2 - C) = \tilde{H}^0(C)$ by Alexander duality, where $C \cong \mathbb{S}^1$, hence $\tilde{H}_0(\mathbb{S}^2 - \mathbb{S}^1) = \mathbb{Z}$ so $H_0(\mathbb{S}^2 - C) = \mathbb{Z} \oplus \mathbb{Z}$. Likewise $H_0(\mathbb{S}^2 - (C - \{x\})) = \mathbb{Z}$ since $C - \{x\} \cong \mathbb{R}$.

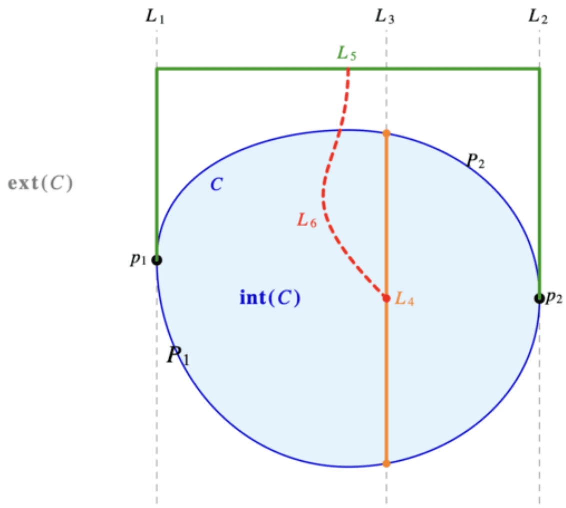
Now, view $\mathbb{S}^2$ as the one-point compactification of $\mathbb{C}$, then $H_0(\mathbb{S}^2 - C) = \mathbb{Z} \oplus \mathbb{Z}$, so $\mathbb{S}^2 - C$ has two connected components, denoted by D_+, D_-.
Since D_+, D_- are disjoint and open, clearly $\partial D_+, \partial D_- \subset C$. Recall the following lemma in general topology: if X, Y disjoint open, P connected and $P \cap X, P \cap Y \neq \emptyset$, then $P \cap \partial X \neq \emptyset$. Analogously, since $H_0(\mathbb{S}^2 - (C - \{x\})) = \mathbb{Z}$, $\mathbb{S}^2 - (C - \{x\})$ is connected, so let $X = D_+, Y = D_-$ and $P = \mathbb{S}^2 - (C - \{x\})$ we obtain $x \in \partial D_+, \partial D_-$ for all $x \in C$. Therefore $\mathbb{S}^2 - C$ has two connected components, with C as their common boundary.
By definition, $\mathbb{S}^2 = \mathbb{C} \cup \{\infty\}$, and we can assume $\infty \in D_+$. Let $U = D_+ - \{\infty\}$ and $V = D_-$, then U, V are open, disjoint, and still connected since ∞ is contained in a disk $O \subset D_+$. Hence U, V are the connected components of $\mathbb{R}^2 - C$, and clearly their boundaries are both C.

Proof using $K_{3,3}$ is not planar **Lemma1**: If C is a PL (Piecewise Linear) closed curve in $\mathbb{R}^2$, then $\mathbb{R}^2 \backslash C$ has exactly two connected components.
Proof: By taking $a \in \mathbb{R}^2 \backslash C$, and for each $x \in \mathbb{R}^2 \backslash C$, consider the parity of number of intersections of C with $[a, x]$ (segment from a to x), if $[a, x]$ passes a vertex or covers an edge, we view it as two intersections. We know that $\mathbb{R}^2 \backslash C$ is disconnected.
If q_1, q_2, q_3 lie in distinct components, take a small disk D such that $D \cap C$ is a line segment. Then

take a tubular neighborhood of C, and walking along this neighborhood we obtain polygonal arcs in $\mathbb{R}^2\backslash C$ sending q_i into D, a contradiction. □

The unbounded component of $\mathbb{R}^2\backslash C$ is $\text{ext}(C)$ and the bounded component is $\text{int}(C)$. Let $\overline{\text{ext}}(C) = C \cup \text{ext}(C)$ and $\overline{\text{int}}(C) = C \cup \text{int}(C)$.
Lemma2: Following Lemma 1, if P is a PL arc in $\overline{\text{int}}(C)$ joining $p, q \in C$ and otherwise disjoint with C. Let P_1, P_2 be the two arcs of C from p to q. Then $\mathbb{R}^2\backslash(C \cup P)$ has 3 components, with boundaries $C, P \cup P_1, P \cup P_2$.
Proof: Clearly $\text{ext}(C)$ is a component of $\mathbb{R}^2\backslash(C \cup P)$, and analogously $\text{int}(C)\backslash P$ has at most two components. Likewise take a segment in $\text{int}(C)$ crossing P, then the two endpoints are in different components by the argument in Lemma1. □
Hence for points x, y on the interior of P_1, P_2, we can't join them without intersecting $C \cup P$.

Theorem3: If C is a simple closed curve in $\mathbb{R}^2$, then $\mathbb{R}^2\backslash C$ is disconnected.
Proof:



Take a disk covering C, and a point outside this disk. Denote the connected component of this point by $\text{ext}(C)$.
Let L_1, L_2 be vertical straight lines intersecting C such that C is entirely in the closed right (resp. left) half plane of L_1 (resp. L_2). Let p_i be the top point on $L_i \cap C$, and P_1, P_2 be the two curves on C connecting p_1, p_2. Let L_3 be a vertical straight line between L_1, L_2. Since $P_1 \cap L_3$ and $P_2 \cap L_3$ are compact and disjoint, L_3 contains an interval L_4 joining P_1, P_2 and having only its ends in C. Let L_5 be a polygonal arc from p_1 to p_2 in $\text{ext}(C)$ consisting of segments of L_1, L_2, and a horizontal straight segment above C. If L_4 is in $\overline{\text{ext}}(C)$, then there is a simple polygonal arc L_6 in $\overline{\text{ext}}(C)$ from L_4 to L_5. But then $C \cup L_4 \cup L_5 \cup L_6$ is a planar representation of $K_{3,3}$, a contradiction. Hence the midpoint of L_4 does not lie in $\overline{\text{ext}}(C)$, so $\text{int}(C)$ is nonempty. □

Lemma4: If G is a 2-connected graph and H is a 2-connected subgraph of G, then G can be obtained from H: in each step add a path of G whose endpoints lie in the current graph.
(This is purely combinatorial, a simple induction on $E(G\backslash H)$.)
Corollary5: If Γ is a planar 2-connected graph with $V(\Gamma) \geqslant 3$ and PL edges, then $\mathbb{R}^2\backslash\Gamma$ has $|E(\Gamma)| - |V(\Gamma)| + 2$ regions, each having a cycle of Γ as boundary.
Proof: Start from a cycle in Γ, and use Lemma4 and Lemma2.

Proposition6: If P is a simple arc (open) in the plane, then $\mathbb{R}^2\backslash P$ is connected.
Proof: Let p, q be two points in $\mathbb{R}^2\backslash P$, and $d > 0$ such that p, q has distance $> 3d$ from P. Since P is the image of a (uniformly) continuous map, we partition P into segments $P_1, \cdots, P_k$ such that P_i joins p_i, p_{i+1}, and each point on P_i has distance $< d$ from p_i.
Let d' be the minimum distance between P_i, P_j for $i \leqslant j-2$, then $d' \leqslant d$.
For each $i = 1, \cdots, k$, partition P_i into segments $P_{i,1}, \cdots, P_{i,k_i}$ such that $P_{i,j}$ joins $p_{i,j}, p_{i,j+1}$ and

that each point on $P_{i,j}$ has distance less than $d'/4$ to $p_{i,j}$, and let Γ_i be the graph consisting of the union of squares with side-length $d'/2$ and $p_{i,j}$ as midpoints. Then we can show both p, q are in the outer face of $\Gamma_1 \cup \cdots \cup \Gamma_k$, since they are outside $B(p_i, 3d)$ while $\Gamma_i \cup \Gamma_{i+1}$ is inside $B(p_i, 3d)$., and P does not intersect that face. Therefore p, q can be joined by a simple polygonal arc disjoint from P. $\square$

If $C \subset \mathbb{R}^2$ is closed and Ω is a region of $\mathbb{R}^2 \backslash C$, then a point p in C is accessible if for some $q \in \Omega$, there is a PL curve from q to p intersecting C only at p.
If P is any arc of C, then Prop6 shows that there is an accessible point on P. Hence the accessible points on C are dense.

Theorem7: (Jordan Curve Theorem) If C is a simple closed curve in $\mathbb{R}^2$, then $\mathbb{R}^2 \backslash C$ has exactly two connected components, having C as boundary.
Proof: Otherwise suppose q_1, q_2, q_3 are points in distinct components $\Omega_1, \Omega_2, \Omega_3$, and let Q_1, Q_2, Q_3 be disjoint segments of C. Ω_i has a simple PL arc $P_{i,j}$ from q_i to Q_j, and we can assume $P_{i,j} \cap P_{i,j'} = \{q_i\}$ for $j \neq j'$. Clearly $P_{i,j} \cap P_{i',j'} = \emptyset$ when $i \neq i'$, so we obtain a planar representation of $K_{3,3}$ (by adding segments in Q_i), a contradiction. Hence $\mathbb{R}^2 \backslash C$ has precisely two components ext(C) and int(C). Since the accessible points are dense, C is the boundary of ext(C) and int(C). $\square$

Likewise we have the generalized version of Lemma2 and Corollary5.

Step II: Schoenflies Theorem

Schoenflies Theorem (1906): $D_+, D_- \cong B(0,1)$. Or a (much) stronger version $\overline{D}_+, \overline{D}_- \cong \overline{B}(0,1)$.
And another version any homeomorphism $f : C \to C'$ can be extended to $\mathbb{S}^2 \to \mathbb{S}^2$.
(This only holds for 2-dim, see Hatcher for Alexander horned sphere.)
(The proof is too complex, omitted here.)

Step III: Any closed surface can be triangulated

Since the surface is compact, we can take a finite covering of local coordinate balls.
In every coordinate ball, we draw a grid, and connect the grids in different balls using PL edges that intersect with the boundaries at finitely many points. We obtain a graph G.
$S \backslash G$ consists of finitely many faces, and by Jordan Schoenflies each face is homeomorphic to a polygon in $\mathbb{R}^2$. The triangulation of polygons is trivial, hence we obtain a triangulation of a closed surface.

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Classification of Closed Surfaces

In this talk, we assume the existence of triangulations of surfaces, and prove the classification theorem for closed surfaces. In Part I, we develop the theory of polygonal presentations, along with a simple example. In Part II, we show the classical proof made by Seifert and Threlfall (1934), which can be found in most textbooks like GTM202. In Part III, we introduce a new proof by Conway (1992), which avoids tedious combinatorial discussions.

Part I: Definitions and Basic Properties

We define polygonal presentations and outline transformations that preserve the homeomorphism class of the geometric realization.
Given a set S, we denote a **word** W in S as a finite list $W = a_1 a_2 \cdots a_n$, where $a_i = s$ or s^{-1} with $s \in S$. A polygonal presentation $\mathcal{P} = \langle S \mid W_1, \cdots, W_n \rangle$ consists of a set S and words W_i in S of length no less than 3 ($\langle a \mid aa \rangle, \langle a \mid aa^{-1} \rangle$ are permitted as a special case).
Any polygonal presentation determines a topological space $|\mathcal{P}|$, called the geometric realization of $\mathcal{P}$, defined as follows:
1. For each word W_i of length k, construct a regular $k-$gon P_i, and mark the k edges using the k letters of W_i bijectively in counterclockwise order.
2. Let $|\mathcal{P}|$ be the quotient space of $\bigsqcup_i W_i$ by pasting edges with the same letter together. Edges marked a are identified preserving orientation, and edges marked a^{-1} are identified reversing orientation.

Consider the following homeomorphism-preserving transformations:

- Reflection: $\langle S \mid a_1 \cdots a_m, W_2, \cdots, W_k \rangle \mapsto \langle S \mid a_m^{-1} \cdots a_1^{-1}, W_2, \cdots, W_k \rangle$.
- Rotation: $\langle S \mid a_1 \cdots a_m, W_2, \cdots, W_k \rangle \mapsto \langle S \mid a_2 \cdots a_m a_1, W_2, \cdots, W_k \rangle$.
- Cutting & Pasting: $\langle S \mid W_1 W_2, W_3, \cdots, W_k \rangle \leftrightarrow \langle S, e \mid W_1 e, e^{-1} W_2, W_3, \cdots, W_k \rangle$. (If W_1, W_2 have length at least 2).
- Folding & Unfolding: $\langle S, e \mid W_1 e e^{-1}, W_2, \cdots, W_k \rangle \leftrightarrow \langle S \mid W_1, \cdots, W_k \rangle$.

Call a pair of edges with the same identifier complementary if they are a, a^{-1} and twisted if they are a, a or a^{-1}, a^{-1} (since the surface twists when one pastes them together).
As an exercise in these operations, we prove a surprising lemma:
Lemma: $K \cong \mathbb{P}^2 \# \mathbb{P}^2$ and $\mathbb{T}^2 \# \mathbb{P}^2 \cong \mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$.
Proof: First show that $K \cong \mathbb{P}^2 \# \mathbb{P}^2$:

$$\begin{aligned} K &= \langle a,b \mid abab^{-1}\rangle \cong \langle a,b,c \mid abc, c^{-1}ab^{-1}\rangle \cong \langle a,b,c \mid bca, a^{-1}cb\rangle \\ &\cong \langle b,c \mid bccb\rangle \cong \langle b \mid bb\rangle \,\#\, \langle c \mid cc\rangle = \mathbb{P}^2 \,\#\, \mathbb{P}^2. \end{aligned}$$

Next $\mathbb{T}^2 \# \mathbb{P}^2 \cong K \# \mathbb{P}^2$:

$$\begin{aligned} K \# \mathbb{P}^2 &= \langle a,b,c \mid abab^{-1}cc\rangle \cong \langle a,b,c,d \mid cabd, d^{-1}ab^{-1}c\rangle \\ &\cong \langle a,b,c,d \mid abdc, c^{-1}ba^{-1}d\rangle \cong \langle a,b,d \mid abdba^{-1}d\rangle \\ &\cong \langle a,b,d,e \mid bde, e^{-1}ba^{-1}da\rangle \cong \langle a,b,d,e \mid deb, b^{-1}ea^{-1}d^{-1}a\rangle \\ &\cong \langle a,d,e \mid deea^{-1}d^{-1}a\rangle \cong \mathbb{T}^2 \# \mathbb{P}^2. \end{aligned} \qquad \square$$

Our goal is to simplify any polygonal presentation into disjoint blocks of $aba^{-1}b^{-1}$ (tori) and cc (projective planes). By this Lemma, any mixture of tori and projective planes reduces entirely to connected sums of projective planes.

Part II: Seifert and Threlfall's Proof

The proof goes by devising algorithms to repeatedly apply the preceding transformations, and simplify the presentation step by step.
Suppose M is a compact connected surface, we assume M has a triangulation, then M has a presentation $\mathcal{P} = \langle S \mid W_1, \cdots, W_n\rangle$ where S are the edges of the triangulation, and W_i are the triangles.

Step1: Reduce to a single face Since M is connected, there must be disjoint faces that share an edge. By repeatedly applying the pasting transformation, we merge all faces into a single polygon. Suppose M is not the sphere $\mathbb{S}^2 \cong \langle a \mid aa^{-1}\rangle$ in the following steps.

Step2: Eliminate adjacent complementary edges Trivial since any sequence aa^{-1} can be eliminated by folding.

Step3: Grouping twisted edges Otherwise if a twisted pair is not adjacent, the word is $VaWa$ where V, W are non-empty. Then

$$\begin{aligned} VaWa &= \langle S,a \mid VaWa\rangle \cong \langle S,a,b \mid Vab, b^{-1}Wa\rangle \cong \langle S,a,b \mid bVa, a^{-1}W^{-1}b\rangle \\ &\cong \langle S,a,b \mid bVW^{-1}b\rangle \cong \langle S,b \mid VW^{-1}bb\rangle. \end{aligned}$$

After this process, the number of nonadjacent pairs decreases by at least one, so after finitely many such operations, there are no nonadjacent twisted pairs. Adjacent complementary pairs that are created can be eliminated by Step2, without introducing new nonadjacent pairs.

Step4: Merge vertices into a single equivalence class Suppose the polygon's vertices fall into multiple equivalence classes. There must be an edge a connecting a vertex from class v to a vertex from class $w \neq v$. Let b be the adjacent edge sharing the v vertex. By Step2 $b \neq a^{-1}$ and clearly $b \neq a$. Since b or b^{-1} must reappear, suppose the presentation is $baXb^{-1}Y$ (the case with $baXbY$ is analogous). Note that

$$\begin{aligned}\langle S, a, b \mid baXb^{-1}Y\rangle &\cong \langle S, a, b, c \mid bac, c^{-1}Xb^{-1}Y\rangle \cong \langle S, a, b, c \mid acb, b^{-1}Yc^{-1}X\rangle \\ &\cong \langle S, a, c \mid acYc^{-1}X\rangle.\end{aligned}$$

Since the edge b, b^{-1} both have vertices v, the process decreased the number of $v-$vertices by at least 1, and added one $w-$vertex. Adjacent complementary pairs created can be eliminated by Step2 without increasing the number of $v-$vertices.
Repeating this eliminates the class v entirely, and eventually there will only be one class.

Step5: Complementary pairs must be intertwined Note that adjacent complementary pairs have been eliminated. Otherwise suppose the presentation is $aXa^{-1}Y$ where X, Y have no edges with common identifiers (since all twisted pairs are adjacent). Then any vertex in X will not be pasted with a vertex in Y, a contradiction to Step4.

Step6: All intertwined pairs are just in the form $aba^{-1}b^{-1}$: Suppose the presentation is $WaXbYa^{-1}Zb^{-1}$, then

$$\begin{aligned}WaXbYa^{-1}Zb^{-1} &= \langle S, a, b \mid WaXbYa^{-1}Zb^{-1}\rangle \cong \langle S, a, b, c \mid WaXc, c^{-1}bYa^{-1}Zb^{-1}\rangle \\ &\cong \langle S, a, b, c \mid XcWa, a^{-1}Zb^{-1}c^{-1}bY\rangle \cong \langle S, b, c \mid XcWZb^{-1}c^{-1}bY\rangle \\ &\cong \langle S, b, c, d \mid WZb^{-1}d, d^{-1}c^{-1}bYXc\rangle \cong \langle S, b, c, d \mid dWZb^{-1}, bYXcd^{-1}c^{-1}\rangle \\ &\cong \langle S, c, d \mid WZYXcd^{-1}c^{-1}d\rangle.\end{aligned}$$

Likewise this process does not split intertwined pairs that are previously adjacent, and makes a new intertwined pair adjacent. Repeating this process for each intertwined pair.

Conclusion We proved that one presentation only contains blocks in the form $aba^{-1}b^{-1}$ or cc, and their identifiers do not repeat by Step2&3. Hence M is either $\mathbb{S}^2$, the connected sum of $\mathbb{T}^2$ or the connected sum of $\mathbb{P}^2$.

Part3: Conway's proof

We now introduce Conway's proof. Recall we proved rather easily that $K \cong \mathbb{P}^2 \# \mathbb{P}^2$ and $\mathbb{T}^2 \# \mathbb{P}^2 \cong \mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$.
We call M an **ordinary surface**, if each component of M is a sphere with perforations (holes), along with attached tori and/or projective planes. Note that we can also attach a Klein bottle since $K \cong \mathbb{P}^2 \# \mathbb{P}^2$.
Define a **zip** to be a closed subset I of ∂M, homeomorphic to $[0, 1]$ or $\mathbb{S}^1$.
We only need to prove the following lemma:

Lemma: Suppose M is an ordinary surface (with boundary), and $I_1, I_2 \subset \partial M$ be disjoint zips. If $f : I_1 \to I_2$ is a homeomorphism, $M' = M/\sim$ is still an ordinary surface where $x \sim f(x)$.
Proof: We discuss the types of I_1, I_2:
Case1: If I_1, I_2 are entire boundary circles, homeomorphic to $\mathbb{S}^1$.
If they belong to different components of M, then glueing them together is just a connected sum, so M' is still ordinary.
Otherwise, pasting I_1, I_2 adds either a handle(#$\mathbb{T}^2$) or a crosscap(#K) depending on whether f is orientation preserving or reversing. Hence M' is still ordinary.
Case2: If I_1, I_2 are arcs that partition the same boundary circle.
Then $I_1 \cup I_2 = C$ is homeomorphic to $\mathbb{S}^1$, and they only intersect at the two endpoints.
If f is orientation reversing, then M' is just M with the perforation removed.
Otherwise f joins I_1, I_2 with antipodal identification. This is just attaching a Möbius band to M, so $M' = M \# \mathbb{P}^2$ is ordinary.
Case3: If I_1, I_2 are arcs that leave gaps in their boundary circles.
First suppose I_1, I_2 are of different boundary circles C_1, C_2. By continuous deformation we may suppose $C_1 \backslash I_1$ and $C_2 \backslash I_2$ are microscopic. Then pasting I_1, I_2 together is just pasting C_1, C_2 together, and removing a tiny open disk formed by $C_1 \backslash I_1$ and $C_2 \backslash I_2$. By Case1 pasting C_1, C_2 makes an ordinary surface, and removing a perforation makes M' still an ordinary surface.
Likewise if I_1, I_2 are arcs of the same boundary circle, we can also suppose they take up almost all of the circle, then it is just pasting the circle as in Case2, and removing one or two open disks. Hence it also leaves M' as an ordinary surface.

Conclusion We assume that any compact manifold M has a triangulation. Make each triangle into a sphere with a hole, where the edges are three arcs that partition the hole. Consider the ordinary surface M', consisting of these spheres, and for each edge in the triangulation, paste the spheres corresponding to the two faces containing the edge. By the lemma, we finally obtain an ordinary surface M'', which is exactly M. Hence if M is connected, it is $\mathbb{S}^2$, the connected sum of $\mathbb{T}^2$, or the connected sum of $\mathbb{P}^2$.

3

C_2-Actions on Closed Surfaces

2026/3/27

See https://arxiv.org/pdf/1612.08489

Target 1: Classifying all free actions on surfaces

(free action: no fixed points) (classify up to equivariant isomorphism)
Theorem: (Classification of free actions)
(a) When g is even there is a unique free C_2-structure on T_g, represented by the antipodal action.
(b) When g is odd, there are two free C_2-structures on T_g: a 180-degree rotation about the central hole, another the antipodal action.
(c) There are no free C_2-structure on N_r when r is odd.
(d) There is exactly one free C_2-structure on N_2, represented by $S^2_a + [DCC]$.
(e) For $s \geqslant 2$ there are exactly two free C_2-structures on N_{2s}, represented by the two C_2-spaces $S^2_a + s[DCC]$ and $T^{anti}_1 + (s-1)[DCC]$.
Here, T_g is the connected sum of g tori, and N_r that of r projective planes.
The space $S^2_a + s[DCC]$ is S^2 under the antipodal action, with s double cross-caps attached to its surface (a cross cap in each half-sphere). Likewise $T^{anti}_1 + s[DCC]$ is the torus with antipodal action, and s double cross caps attached.

We admit the following proposition:
There are bijections

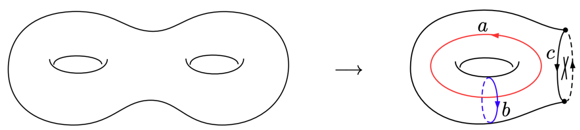
$$(\mathbb{Z}/2 - \text{principal bundle over } Y) \longleftrightarrow [Y, B\mathbb{Z}/2] \longleftrightarrow H^1(Y, \mathbb{Z}/2).$$

Here, for a discrete group G, $G-$principal bundle over Y means regular covering spaces of Y with Deck group G.
BG is defined by the universal property: Take EG the contractible topological space with free $G-$action, and let BG be the orbits EG/G.
$[Y, BG]$ is the homotopy classes of continuous maps $Y \to BG$.
$H^1(Y, \mathbb{Z}/2) \cong \mathrm{Hom}(H_1(Y, \mathbb{Z}/2), \mathbb{Z}/2)$.

Def: For N_r and T_g we define $S(N_r)$ and $S(T_g)$ to be the set of isomorphism classes of surfaces with C_2-action such that the quotient is N_r (or T_g).
Prop: There is a bijection

$$S(Y) \longleftrightarrow H^1(Y, \mathbb{Z}/2)/\mathrm{Aut}(Y).$$

Example: Consider T_2 with antipodal action. The quotient space is a torus with a cross cap, which is $T_2/C_2 \cong N_3$, and $H_1(N_3; \mathbb{Z}_2)$ is generated by the elements a, b, c. Both a and b lift to loops under the projection $T_2 \to T_2/C_2$, whereas c does not. So under the bijection $S(Y) \to H^1(Y, \mathbb{Z}/2)/\mathrm{Aut}(Y)$ the C_2-space T_2 corresponds to the map $H_1(N_3; \mathbb{Z}_2) \to \mathbb{Z}/2$ sending $a \mapsto 0, b \mapsto 0, c \mapsto 1$.



Calculate that $H^1(T_g, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^{2g}$, $H^1(N_r, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^r$.
So we should consider the image $\mathrm{Aut}(Y) \to GL_{2g}(\mathbb{Z}/2)$.
Fact: There is a product $H^1(Y, \mathbb{Z}/2) \times H^1(Y, \mathbb{Z}/2) \to \mathbb{Z}/2$. And $\mathrm{Aut}(T_g) \to \mathrm{Sp}(2g, \mathbb{Z}/2) \to \mathrm{GL}(2g, \mathbb{Z}/2)$, where $\mathrm{Sp}(2g, \mathbb{Z}/2)$ is the symplectic group. And $\mathrm{Aut}(N_r) \to O(r, \mathbb{Z}/2) \to \mathrm{GL}(r, \mathbb{Z}/2)$.
Injectivity is trivial. Now we prove $M(T_g) \to Sp(2g, \mathbb{Z}/2)$ and $M(N_r) \to O(r, \mathbb{Z}/2)$ are surjections.
Step1: Find a set of generators of $O(r, \mathbb{Z}/2)$ & $Sp(2g, \mathbb{Z}/2)$.
Lemma1 (Coro4.14): Let $n \geqslant 1$, then $O(n, \mathbb{Z}/2)$ is generated by the permutation matrices together with (if $n \geqslant 4$) the single matrix $A \oplus I_{n-4}$ where $A = 1_{n\times n} - I$.
Proof: We use induction on r. Assume $M \in O(r)\backslash H$, where H generated by permutation and $A \oplus I_{n-4}$, by induction assumption, $Me_i \neq e_i$ for some $i = 1, \cdots, r$. Note that H acts transitively on $v \in \mathbb{F}_2^r - \Omega$ with $\langle v, v\rangle = 1$, where $\Omega = (1, \cdots, 1)$. Note that $\langle Me_1, Me_1\rangle = \langle e_1, e_1\rangle = 1$, and $M\Omega = \Omega$ so $Me_1 \neq \Omega$. Let $Me_1 = w \neq \Omega$ and take $B \in H$ such that $Bw = e_1$, then $BMe_1 = e_1$ implies $BM \in H$, so $M \in H$, a contradiction.
Step2: Find the preimage of permutations and A.
Def: (Dehn twist along loop) Given a smooth loop $l : \mathbb{S}^1 \to X$, take the tubular neighborhood of l, denoted $\tilde{l} : \mathbb{S}^1 \times [0, 1] \xrightarrow{f} X$, then there is an automorphism of $\tilde{l}$ given by $D(\tilde{l}) : \mathbb{S}^1 \times [0, 1] \to X, (t, x) \mapsto f(e^{2\pi i x} t, x)$.
For permutations: Choose a loop connecting two crosscap and use the associated Dehn twist, $c_1 \mapsto c_1 + c_1 + c_2 = c_2$ and $c_2 \mapsto c_1$. Likewise for A we take the loop connecting four crosscaps.
Step3: Consider the orbits of $(\mathbb{Z}/2)^{2g}/\mathrm{Sp}(2g, \mathbb{Z}/2)$ and $(\mathbb{Z}/2)^r/O(r, \mathbb{Z}/2)$.
For $O(r, \mathbb{Z}/2)$ there are four orbits, namely $0 = (0, \cdots, 0), a_1 = (1, 0, \cdots, 0)$, $a_2 = (1, 1, 0, \cdots, 0)$ and $\Omega = (1, 1, \cdots, 1)$.
For $\mathrm{Sp}(2g, \mathbb{Z}/2)$ there are two orbits:
Target: All non-zero elements of $(\mathbb{Z}/2)^{2g}$ are in the same orbit.
(1) use induction, verify $g = 1$. (2) choose a standard basis of $(\mathbb{Z}/2)^{2g}$ such that the inner product is given by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus \cdots \oplus \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which naturally induces $\mathrm{Sp}(2) \times \cdots \times \mathrm{Sp}(2) \to \mathrm{Sp}(2g)$. By the case $g = 1$ there are at most 2^g orbits $[B_1, \cdots, B_g]$ where $B_i \in \{O, T\}$ for $O = [0, 0]$ and $T = [1, 0]$. Since permuting the symplectic blocks are all elements of $\mathrm{Sp}(2g, \mathbb{Z}/2)$, we can reduce to at most $g + 1$ orbits $[O, \cdots, O], [T, O, \cdots, O], \cdots, [T, \cdots, T]$. Consider the symplectic matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

then $A[T, O] = [T, T]$, so applying A we infer $[T, O, \cdots, O], \cdots, [T, \cdots, T]$ are of the same orbit.

Using the statements above, we can classify all C_2-actions.

Target2: Classify C_2-actions on orientable spaces

Main Theorem: The number of isomorphism classes of C_2-actions on T_g is equal to $4+2g$.

Lemma: $X - X^{C_2}$ has at most 2 connected components.

Prop: Suppose an orientation preserving involution on T_g has finite fixed set of size n, then $n \leqslant 2+2g$ and $n \equiv 2+2g \pmod 4$. (Easy using Euler characteristics)
Prop: (Harnack's Theorem) Let σ be an orientation reversing action on T_g, and C the number of ovals in the fixed set. If the action is separating then $C \leqslant 1+g$ and $C \equiv 1+g \pmod 2$. If it is non-separating (connected after action) then $C \leqslant g$.

Lemma: Up to isomorphism, the only C_2-actions on S^2 are $S^{2,0}$ (trivial), $S^{2,1}$ (reflection across equatorial plane), $S^{2,2}$ (180degree rotation around an axis) and S^2_a (antipodal).

4

Riemann Surfaces

Reference: GTM81 Riemann surfaces, Forster; Complex Analysis, Stein.
Building upon our previous classification of closed surfaces, in this talk, we explore the classification of Riemann surfaces through two foundational theorems, both first developed by Riemann:

- **Uniformization Theorem**: (Analytical) Every simply connected Riemann surface is biholomorphic to either the complex sphere $\mathbb{P}^1$, the complex plane $\mathbb{C}$, or the unit disk $\mathbb{D}$.
- **Projective Embedding**: (Algebraic) Every compact Riemann surface can be embedded into projective space. Furthermore, the category of compact Riemann surfaces, smooth algebraic curves, and function fields of transcendental degree 1 are equivalent.

Preliminaries

A Riemann surface X is a (connected) topological manifold, such that for every pair of charts $(\varphi_\alpha, U_\alpha), (\varphi_\beta, U_\beta)$, $\varphi_\alpha \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$ is holomorphic. A function $f : X \to Y$ is holomorphic if for every local chart $(\varphi_\alpha, U_\alpha) \subset X, (\varphi_\beta, U_\beta) \subset Y$, $\varphi_\alpha f \varphi_\beta^{-1}$ is holomorphic in the usual sense.
Examples of compact Riemann surfaces:

- The Riemann surface of genus $g = 0$ is the sphere $\mathbb{P}^1$.
- Riemann surfaces of genus $g = 1$ are complex tori $\mathbb{C}/\Gamma$ where $\Gamma = \omega_1\mathbb{Z} + \omega_2\mathbb{Z}$, characterized by $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$, or the $j-$invariant $j = 1728\frac{4a^3}{4a^3+27b^2}$ for $y^2 = x^3 + ax + b$.

$f : X \to \mathbb{C}$ is meromorphic if $f : X \to \mathbb{P}^1$ is holomorphic. $f : X \to Y$ is biholomorphic if it has a holomorphic inverse.
A holomorphic/meromorphic 1-form ω is a 1-form that is locally in the form $\omega = f dz$ where f holomorphic/meromorphic. Let $\mathcal{M}(X)$ be the set of meromorphic functions $f : X \to \mathbb{C}$, and $\mathcal{M}^{(1)}(X)$ meromorphic 1-forms.

Projective Embedding

Suppose X is a compact Riemann surface with genus g.
Definition: A **divisor** on X is a mapping $D : X \to \mathbb{Z}$ such that for any compact $K \subset X$, $D(x) \neq 0$ for only finite $x \in K$. (Here, its just $D(x) \neq 0$ for finite $x \in X$.) The set of divisors form an Abelian group $\mathrm{Div}(X)$, partially ordered by $D \leqslant D' \iff D(x) \leqslant D'(x) \forall x \in X$.
Definition: For open $Y \subset X$, $f \in \mathcal{M}(Y)$ and $a \in Y$, define

$$\mathrm{ord}_a(f) = \begin{cases} 0, & f \text{ holomorphic and nonzero at } a \\ k, & f \text{ has a zero of order } k \\ -k, & f \text{ has a pole of order } k \\ \infty, & f \text{ vanishes locally} \end{cases}$$

Then $x \mapsto \mathrm{ord}_x(f)$ is a divisor, denoted (f) and called a principal divisor.
Definition: If X is compact, one can define a homomorphism

$$\deg : \mathrm{Div}(X) \to \mathbb{Z}, D \mapsto \sum_{x \in X} D(x).$$

Then $\deg(f) = 0$ for any $f \in \mathcal{M}(X)$. For a meromorphic 1-form ω and $a \in X$, suppose locally $\omega = g dz$, then we define $\mathrm{ord}_a(\omega) = \mathrm{ord}_a(g)$. Since every pair of 1-forms ω_1, ω_2 satisfy $\omega_1 = f\omega_2$, $(\omega_1) = (\omega_2) + (f)$, so the equivalence class $[K] = [(\omega)]$ is independent of the choice of ω.

Theorem (Riemann-Roch): Suppose D is a divisor on a compact Riemann surface X of genus g, then $H^0(X, \mathcal{O}_D)$ and $H^1(X, \mathcal{O}_D)$ are finite dimensional vector spaces, and

$$\dim H^0(X, \mathcal{O}_D) - \dim H^1(X, \mathcal{O}_D) = 1 - g + \deg D.$$

Here, $H^0(X, \mathcal{O}_D) = L(D) = \{f \in \mathcal{M}(X) : (f) \geqslant -D\}$.
By Serre duality, we know

$$\dim H^1(X, \mathcal{O}_D) = \dim H^0(X, \mathcal{O}_{K-D}),$$

hence Riemann-Roch can be written as

$$\dim L(D) - \dim L(K - D) = 1 - g + \deg D.$$

As a corollary, let $D = 0, K$ we obtain $\deg K = 2g - 2$. This also gives another definition of genus: $g = \dim L(K)$.
Using Riemann-Roch, we can **classify compact Riemann surfaces of genus zero**:
Suppose X is a compact Riemann surface of genus g, and $a \in X$, then set $D(a) = g+1$ and $D(x) = 0$ otherwise, we obtain $\dim L(D) \geqslant 1 - g + \deg D = 2$ so there is a non-constant $f \in L(D)$. Then f is meromorphic, with a pole of order $\leqslant g + 1$ at a, and otherwise holomorphic, so f is a holomorphic covering map $f : X \to \mathbb{P}^1$ with at most $g + 1$ sheets. When $g = 0$, f is a biholomorphism so X is isomorphic to $\mathbb{P}^1$.

Let D be a divisor on X, we call D (the sheaf $\mathcal{O}_D$) globally generated if for any x there exists $f \in L(D)$ such that $\mathrm{ord}_x(f) = -D(x)$.
Theorem: If X has genus g, and $\deg D \geqslant 2g$, then D is globally generated.
Proof: Suppose $x \in X$ is a fixed point, let $D' = D - x$ ($D'(y) = D(y) \forall y \neq x$ and $D'(x) = D(x) - 1$). Since $\deg D > \deg D' > 2g - 2$, $\deg(K - D') < 0$ so we have $L(K - D) = L(K - D') = 0$, and by Riemann-Roch $\dim L(D) > \dim L(D')$. Take $f \in L(D) \backslash L(D')$, then $\mathrm{ord}_x(f) = -D(x)$.

For a compact surface X of genus $g > 0$, let D be a divisor of degree $\geqslant 2g+1$. Take a basis $f_0, \cdots, f_N$ of $L(D)$, we construct an embedding $F = (f_0 : \cdots : f_N) : X \to \mathbb{P}^N$:
(i) Definition: For any $x \in X$, take a neighborhood (V, z) where $z(x) = 0$, and let $k = \min \mathrm{ord}_x(f_i)$. Suppose $f_i(z) = z^k g_i(z)$ then g_i are holomorphic and $g_i(x)$ not identically zero, and we can define $F(x) = (g_0(x) : \cdots : g_N(x))$. Suppose $g_j(x) \neq 0$, then locally F can be viewed as $F : X \to \mathbb{C}^N$, $F = \left(\frac{g_0}{g_j}, \cdots, \frac{g_{j-1}}{g_j}, \frac{g_{j+1}}{g_j}, \cdots, \frac{g_N}{g_j}\right)$, so F is holomorphic.

(ii) Injective: Suppose $x_1 \neq x_2$ in X, take $D' = D - x_2$, then deg $D' = 2g$ so D' is globally generated. Take $f \in L(D')$ such that $\text{ord}_{x_1}(f) = -D(x_1)$, and $\text{ord}_{x_2}(f) > -D(x_2)$. Suppose $f = \sum \lambda_i f_i$, and take local coordinates $(V_1, z_1), (V_2, z_2)$ such that $z_i(x_i) = 0$. Suppose $k_j = \min \text{ord}_{x_j}(f_i)$, and $f_i(z_j) = z_j^{k_j} g_{i,j}(z)$. Then by construction of f, $g_1(x_1) \neq 0$ while $g_2(x_2) = 0$, so $F(x_1) \neq F(x_2)$ and F is injective.
(iii) Immersion: Let $x_0 \in X$ and $D' = D - x_0$. Take $f \in L(D')$ such that $\text{ord}_{x_0}(f) = -D(x_0) + 1$. Suppose $f = \sum \lambda_i f_i$, and let (V, z) be a neighborhood such that $f = z^k g, f_i = z^k g_i$ where $k = \min \text{ord}_{x_0}(f) = -D(x_0)$. WLOG assume $g_0(x_0) \neq 0$. Then $F_0 = \varphi_0 \circ f = (F_{01}, \cdots, F_{0N}) : V \to \mathbb{C}^N$ is given by $F_0 = \left(\frac{g_1}{g_0}, \cdots, \frac{g_N}{g_0}\right)$, and $\sum \lambda_i dF_{0i} = d(g/g_0)$. Since $g_0(x_0) \neq 0$ and g has a simple zero at x_0, $d(g/g_0) \neq 0$ so at least one $dF_{0i} \neq 0$.

Example: If $X = \mathbb{C}/\Gamma$, $\Gamma = \mathbb{Z} + \tau\mathbb{Z}$, then $g = 1$, so we have an embedding $X \to \mathbb{P}^2, \bar{z} \mapsto (\wp(z) : \wp'(z) : 1)$, with the relation $(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3)$. (This is injective since $\wp$ is even with order 2.)

Recall that given homogeneous polynomials $\{F_\alpha\}$, an algebraic variety $C \subset \mathbb{P}^n$ is $\{p : F_\alpha(p) = 0 \forall \alpha\}$. Algebraic curves are varieties of dimension 1, and an analytic variety is locally the zero set of a holomorphic function.
Chow's Theorem: Any analytic subvariety of projective space is algebraic. Holomorphic maps between analytic subvarieties induce regular rational morphisms between algebraic varieties.
Therefore, every compact Riemann surface is isomorphic to a smooth algebraic curve.

Recall the classic theorem that a topological space X is completely determined by $C(X)$, namely $X \cong \text{Max}(C(X))$. There is a similar result for Riemann surfaces:
For a compact Riemann surface X, $\mathcal{M}(X) \cong \text{Frac}(\mathbb{C}[x,y]/P(x,y))$ is a function field of transcendental degree 1, and X is determined by the DVRs of $\mathcal{M}(X)$ containing $\mathbb{C}$. Any homomorphism $f : X \to Y$ gives a $\mathbb{C}-$algebra hom $f^* : \mathcal{M}(Y) \to \mathcal{M}(X)$.

Therefore, we obtain a three-fold categorical equivalence:
Compact Riemann surfaces - Smooth algebraic curves - Function fields over $\mathbb{C}$ with transcendental degree 1.
And objects like point-DVR or Deck transformation-Galois group correspond to each other. Hence we can study the same objects using tools from both analysis and algebra.
If X compact, $X \to \mathcal{M}(X)$ and $f : X \to Y \implies f^* : \mathcal{M}(Y) \to \mathcal{M}(X)$.
If F field over $\mathbb{C}$ of transcendental degree 1, $X = \{\mathcal{O} \supset \mathbb{C} : \mathcal{O} \text{ DVR}\}$, and the local chart at $\mathcal{O}$ is $t \in F$ which generates $\mathfrak{m}_{\mathcal{O}}$.

Example: $Y = \mathbb{P}^1$, then $F_Y = \mathcal{M}(Y) \cong \mathbb{C}(z)$. Point $a \in Y$ -> DVR $\mathcal{O}_a = \{f \in \mathbb{C}(z) : \text{ord}_a(f) \geqslant 0\}$ with max ideal $\mathfrak{m}_a = \{f : f(a) = 0\}$. $\text{Aut}(Y) = \text{PSL}(2, \mathbb{C})$ -> $\text{Gal}(\mathbb{C}(z)/\mathbb{C}) = \text{PSL}(2, \mathbb{C})$.
Example: $X = \mathbb{C}/\Gamma, \Gamma = \mathbb{Z} + i\mathbb{Z}$ -> elliptic curve $y^2 = x^3 - x$ -> field extension $F_X = \text{Frac}(\mathbb{C}[x,y]/(y^2 - x^3 + x))$. The covering map $\wp : X \to Y = \mathbb{P}^1$ is even and has double poles, giving $\text{Deck}(X/Y) = \mathbb{Z}/2\mathbb{Z}$ -> $\text{Gal}(F_X/F_Y) = \mathbb{Z}/2\mathbb{Z}$ by $y \mapsto -y$.

Uniformization Theorem

Main Theorem: If X is a simply connected Riemann surface, then X can be biholomorphically mapped onto either the Riemann sphere $\mathbb{P}^1$, the complex plane $\mathbb{C}$, or the unit disk $\mathbb{D}$.

Recall the proof of Riemann mapping theorem, we first show simply connected open sets are biholomorphic to $\mathbb{D}$, then solve the Dirichlet problem using the biholomorphism. Here, the process is reversed.

Lemma1: Suppose X is a non-compact Riemann surface, and $Y \subset X$ is simply connected with ∂Y piece-wise smooth. Then there exists a biholomorphic mapping of Y onto the unit disk $\mathbb{D}$.
(This is PDE, using harmonic analysis and Weyl's lemma to solve the Dirichlet problem.)
Koebe-Bieberach Theorem: If $f : \mathbb{D} \to \mathbb{C}$ holomorphic and injective, $f(0) = 0, f'(0) = 1$, then

$D\left(\frac{1}{4}\right) \subset f(\mathbb{D})$, and $|f(z)| \leqslant |z|/(1-|z|)^2$. See Stein, Problem3.1.

Definition: A family $\mathcal{F}$ of holomorphic functions is called normal if every sequence in $\mathcal{F}$ has subsequence converging uniformly on every compact subset (the limit may not be in $\mathcal{F}$).
Montel's theorem: Recall Arzelá-Ascoli, locally uniform boundedness & equicontinuous implies $\mathcal{F}$ is a normal family. By the Cauchy integral formula uniform boundedness implies equicontinuous.
Hurwitz's theorem: Note that if f_n is injective and converges uniformly to f on every compact subset, then f is injective or constant, using the Cauchy integral formula.

Proof of main theorem:
(i) If X is compact, then X is of genus $g=0$ since it is an orientable & simply connected closed surface (using classification of closed surfaces), so X is the Riemann sphere $\mathbb{P}^1$.
(ii) Now assume X is not compact.
Take an exhaustion of connected compact sets $K_0 \subset K_1 \subset \cdots$. For each n, find a relatively compact open neighborhood $V_n \supset K_n$ such that ∂V_n is piece-wise smooth (union of coordinate disks). Let Y_n be the union of V_n and compact components of $X \backslash V_n$. Since X is simply connected, so are Y_n hence by Lemma1 each Y_n is biholomorphic to $\mathbb{D}$.
Choose a point $a \in Y_0$ and a coordinate neighborhood (U, z) of a, then there exists a real number $r_n > 0$ and a biholomorphic mapping $f_n : Y_n \to D(r_n)$ with $f_n(a) = 0$ and $f_n'(a) = 1$.
(iii) Note that $r_n \leqslant r_{n+1}$ for all n, since $h = f_{n+1} \circ f_n^{-1} : D(r_n) \to D(r_{n+1})$ satisfies $h(0) = 0$ and $h'(0) = 1$, so by Schwarz lemma $1 \leqslant r_{n+1}/r_n$. Let $R = \lim_{n\to\infty} r_n \in (0, \infty]$. We show that X is biholomorphic to $D(R)$ which is either $\mathbb{D}$ or $\mathbb{C}$.
Let $h_{m,n} = f_m f_n^{-1}$ then its normalization $g(x) = h_{m,n}(x r_n)/r_n$ is injective and $g(0) = 0, g'(0) = 1$.
(iv) We show that $\{f_n\}$ is normal, and obtain $f : X \to D(R)$.
Case1: $R < \infty$, then f_n is bounded by R.
Case2: $R = \infty$. By Koebe distortion theorem $|g(z)| \leqslant |z|/(1-|z|)^2$, so for $w \in D(r_n)$, $|h_{m,n}(\omega)| \leqslant r_n^2|\omega|/(r_n - |\omega|)^2$. So $h_{m,n}$ and also f_n are locally uniformly bounded.
By Montel's theorem $\{f_n\}$ has a subsequence that converges uniformly on any compact subset, and denote the limit by f. Since f_n are injective, f is injective, $f(a) = 0$ and $f'(a) = 1$.
(v) Finally we show that $f : X \to D(R)$ is surjective:
Case1: $R = \infty$. Consider $g : \mathbb{D} \to \mathbb{C}$ of $h_{m,n}$, then by Koebe 1/4 theorem $D(1/4) \subset g(\mathbb{D})$. Hence $D(r_n/4) \subset h_{m,n}(D(r_n)) = f_m(Y_n)$. Let $m_k \to \infty$ we obtain $f(X) \supset f(Y_n) \supset D(r_n/4)$, so $f(X) \supset \bigcup_{n=1}^{\infty} D(r_n/4) = \mathbb{C}$. Therefore $f(X) = \mathbb{C}$.
Case2: $R < \infty$. Suppose the contrary, we find a biholomorphic map $g : f(X) \to D(r)$ with $r < R$ such that $g(0) = 0$ and $g'(0) = 1$. Take $r_n > r$, then $h = g \circ f \circ f_n^{-1} : D(r_n) \to D(r)$ satisfy $h(0) = 0, h'(0) = 1$ so it contradicts Schwarz lemma.
Let $Y = f(X)$ be a proper simply connected subset of $D(R)$, we can suppose $R = 1$. Take $a \in \mathbb{D}\backslash Y$, and $\psi_a = \frac{z-a}{1-z\bar{a}} \in \mathrm{Aut}(\mathbb{D})$. Since $\psi_a|_Y \neq 0$, take $g : Y \to \mathbb{D}$ such that $g^2 = \psi_a|_Y$. Let $b = g(0)$ and take ψ_b, then $h = \psi_b \circ g$ satisfy $h(0) = 0, h'(0) = \gamma = (1+|b|^2)/2b$. Hence $h/\gamma : Y \to D(1/|\gamma|)$ has the desired properties.
(Same as Stein's proof of the Riemann mapping theorem.)

Applications: For a Riemann surface X, if the universal covering $\tilde{X}$ is $\mathbb{P}^1/\mathbb{C}/\mathbb{D}$, we call it elliptic/-parabolic/hyperbolic.
Recall that X can be viewed as $\tilde{X}/G$ where $G = \mathrm{Deck}(\tilde{X}/X)$ acts freely and discretely.
Proposition: (a) Every automorphism of $\mathbb{P}^1$ has a fixed point.
(b) If $G < \mathrm{Aut}(\mathbb{C})$ which acts discretely and freely, $G = \langle id \rangle$, $G = \langle \tau_\gamma \rangle$ where $\tau_\gamma : z \mapsto z + \gamma$, or $G = \langle \tau_{\gamma_1}, \tau_{\gamma_2} \rangle$ where γ_1, γ_2 are independent over $\mathbb{R}$.
Easy to verify since automorphism of $\mathbb{P}^1$ and $\mathbb{C}$ are in the form $\frac{az+b}{cz+d}$ and $az + b$.
Corollary: (a) $\mathbb{P}^1$ is elliptic; (b) $\mathbb{C}, \mathbb{C}^*, \mathbb{C}/\Gamma$ are parabolic; (c) all other Riemann surfaces are hyperbolic.
We obtain Picard's little theorem as an easy corollary:
Corollary: (Picard's little theorem) Suppose $f : \mathbb{C} \to \mathbb{C}$ is non-constant and entire, then f takes every value of $\mathbb{C}$ with at most one exception.

5

Introduction to Morse Theory

2026/4/10

Milnor *Morse Theory* Chapters 1,2,3.1

Intro: What is Morse Theory?

Let M be a differential surface, $f : M \to \mathbb{R}$ a smooth function (called height function). Define $M^a = \{x \in M : f(x) < a\}$ or $\leqslant a$ if M has boundary.
e.g. M is a torus, f the height of a point which has 4 critical points, then M^a and its homology class at critical points are:
$\emptyset \cong \emptyset$, cone = point (0-cell), handle = circle (attach 1-cell), torus minus circle = figure eight (attach 1-cell), torus = torus (attach 2-cell).

Basic Theory

Definition: For f and $p \in M$, p is a **critical point** if df_p is zero.
For a critical point p, define the **Hessian** $H(f)$ at p: For $v, w \in T_pM$, with extensions $\tilde{v}, \tilde{w}$ to vector fields, define $H(f)(v,w) = \tilde{v}_p(\tilde{w}(f))$. Verify that $H(f)$ is a bilinear form, and it is symmetric since

$$\tilde{v}_p(\tilde{w}(f)) - \tilde{w}_p(\tilde{v}(f)) = [\tilde{v}, \tilde{w}]_p(f) = df_p([\tilde{v},\tilde{w}]_p) = 0$$

by p is a critical point of f. It is well-defined since $\tilde{v}_p = v_p$ so H is independent of $\tilde{v}$, and by symmetry is independent of $\tilde{u}$.
In local coordinates, the Hessian becomes the standard hessian $\left(\frac{\partial^2 f_i}{\partial x^i \partial x^j}\right)_{i,j}$. What we've shown is that the Hessian is independent of the choice of coordinates, if p is a critical point.
The **index** of $H(f)$ is the maximal dimension of a subspace of T_pM such that $H(f)$ is negative definite.
The **nullity** of $H(f)$ is $\dim\{v : H(f)(v,w) = 0 \forall w\}$.

Morse Lemma For non-degenerate p (nullity=0), there exists local coordinates $(y^1, \cdots, y^n)$ in a neighborhood U, such that $y(p) = 0$, and $f = f(p) - (y^1)^2 - \cdots - (y^\lambda)^2 + (y^{\lambda+1})^2 + \cdots + (y^n)^2$, where λ is the index.

Lemma (Hadamard): U is a convex neighborhood of $0 \in \mathbb{R}^n$, $f : U \to \mathbb{R}$ is C^∞ and $f(0) = 0$, then $f(x_1, \cdots, x_n) = \sum_{i=1}^n x_i g_i(x_1, \cdots, x_n)$, where $g_i \in C^\infty(U)$ and $\frac{\partial f}{\partial x_i}(0) = g_i(0)$.

Proof: By Hadamard's lemma, we have $f(x_1, \cdots, x_n) = \int_0^1 \frac{df(tx_1,\cdots,tx_n)}{dt}\, dt$ where $\frac{df(tx_1,\cdots,tx_n)}{dt} = \sum \frac{\partial f}{\partial x_i} x_i$. Hence just take $g_i(x) = \int_0^1 \frac{\partial f}{\partial x_i}(tx)\, dt$.

Proof of Morse Lemma: Apply the lemma twice, we obtain $f = \sum_{i,j} x_i x_j h_{i,j}(x_1, \cdots, x_n)$, WLOG assume $h_{i,j} = h_{j,i}$, then $h_{ij}(0) = \frac{1}{2}\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f|_{(0,0)}$. Use induction, and suppose we already have $\varphi : U_{r-1} \to \mathbb{R}^n$ smooth bijective and $f \circ \varphi(u) = \pm(u_1)^2 \pm \cdots \pm (u_{r-1})^2 + \sum_{i,j \geqslant r} u_i u_j H_{ij}(u_1, \cdots, u_n)$ in a neighborhood U_{r-1} (of the new coordinate), such that $\det\left(\frac{\partial \varphi_i}{\partial u_j}\right)_{1\leqslant i,j\leqslant m}|_{u=0} \neq 0$. Then if we set $H_{ij} = H_{ji}$, at least one $H_{ij}(0) \neq 0$, and we can take a suitable φ such that $H_{rr}(0) \neq 0$.
Take a neighborhood U_r such that $g(\mathbf{u}) = \sqrt{H_{rr}(\mathbf{u})}$ is smooth. Take $v_i = u_i \forall i \neq r$, and $v_r(\mathbf{u}) = g(\mathbf{u})\left(u_r + \sum_{i>r} u_i \frac{H_{ir}(\mathbf{u})}{H_{rr}(\mathbf{u})}\right)$. Then in U_r with the new coordinates, $f \circ \varphi \circ \psi^{-1}(v) = \pm(v_1)^2 \pm \cdots \pm (v_r)^2 + \sum_{i,j>r} v_i v_j H'_{ij}(\mathbf{v})$ and $\det\left(\frac{\partial \psi_i}{\partial u_j}\right)_{1\leqslant i,j\leqslant m}|_{u=0} \neq 0$. Hence by induction there is a smooth bijection $g : V \to \mathbb{R}^n$ such that $\det\left(\frac{\partial g_i}{\partial y_j}\right) \neq 0$ and $f \circ g(y) = f(0) - y_1^2 - \cdots - y_k^2 + y_{k+1}^2 + \cdots + y_m^2$. By definition, λ will be the index of the Hessian. $\square$

Corollary: Non-degenerate critical points are isolated.

Homotopy Type in Terms of Critical Values

In this section $M^a = f^{-1}((-\infty, a])$ so it is closed.
Definition: A 1-parameter group of diffeomorphisms of a manifold M is a C^∞ map $\varphi : \mathbb{R} \times M \to M$ such that:
1. $\varphi_t = \varphi(t, \cdot)$ is a diffeomorphism on M.
2. $\varphi_{t+s} = \varphi_t \circ \varphi_s$ for all $t, s \in \mathbb{R}$.

Given φ, we can define a vector field $X_q(f) = \lim_{h\to 0} \frac{f(\varphi_h(q)) - f(q)}{h}$. We say X generate φ.
Lemma: If X is a smooth manifold M with compact support, then X generates a unique 1-parameter group of diffeomorphisms. (Use Picard's theorem for linear ODE's).

Theorem3.1: $f \in C^\infty(M)$, $a < b$, $f^{-1}([a, b])$ compact without critical points, then M^a is isomorphic to M^b under diffeomorphism, furthermore M^a is a deformation retract of M^b so $M^a \hookrightarrow M^b$ is a homotopy equivalence.
Proof: Choose a Riemannian metric g on M. The gradient of f is a vector field $\langle X, \mathrm{grad}(f)\rangle = X(f)$ for any vector field X. Take smooth and compactly supported $\rho : M \to \mathbb{R}$ such that $\rho = 1/\langle \mathrm{grad} f, \mathrm{grad} f\rangle$ on $f^{-1}([a, b])$. Consider the vector field $X : X_q = \rho(q)(\mathrm{grad} f)_q$, then it generates a 1-parameter group of diffeomorphisms $\varphi_t : M \to M$.
Consider $t \mapsto f(\varphi_q(t))$, then $\frac{d(f(\varphi_q(t)))}{dt} = \left\langle \frac{d\varphi_t(q)}{dt}, \mathrm{grad} f \right\rangle = \langle X, \mathrm{grad} f\rangle = 1$. Hence $t \mapsto f(\varphi_q(t))$ is linear, i.e. $f(\varphi_q(t)) = f(q) + t$ as long as $f(\varphi_q(t)) \in [a, b]$. Therefore φ_{b-a} gives the diffeomorphism $M^a \simeq M^b$.
Define $r_t : M^b \to M^a$ by $r_t(q) = \begin{cases} q, & f(q) \leqslant a \\ \varphi_{t(a-f(q))}(q), & a \leqslant f(q) \leqslant b \end{cases}$, then $r_0 = id$ and r_1 is a retraction $M^b \to M^a$. Hence M^a is a deformation retraction of M^b. $\square$

Theorem3.2: Let $f : M \to \mathbb{R}$ be smooth, p a non-degenerate critical point with index λ. Let $f(p) = c$, suppose for some $\varepsilon > 0$, $f^{-1}([c - \varepsilon, c + \varepsilon])$ is compact, and contains no critical point of f other than p. Then for all sufficiently small ε, $M^{c+\varepsilon}$ has the homotopy type of $M^{c-\varepsilon}$ with a $\lambda-$cell attached.
Proof: By Morse lemma, there is a local coordinate (U, u^i) such that $f = c - (u^1)^2 - \cdots - (u^\lambda)^2 + (u^{\lambda+1})^2 + \cdots + (u^n)^2$. Take $\varepsilon > 0$ small such that $f^{-1}([c-\varepsilon, c+\varepsilon])$ is compact and has only one critical point p, and the image of U under the coordinate embedding contains the ball $\sum_{i=1}^n (u^i)^2 \leqslant 2\varepsilon$.
Step1: Construct $F : M \to \mathbb{R}$ such that $F \leqslant f$ and $F = f$ away from p.
Choose $\mu : \mathbb{R} \to \mathbb{R}$, $\mu(0) > \varepsilon$, $\mu(r) = 0 \forall r \geqslant 2\varepsilon$ and $-1 < \mu' \leqslant 0 \forall r$.

Let $F = f - \mu(\xi + 2\eta)$, where $\xi = (u^1)^2 + \cdots + (u^\lambda)^2$ and $\eta = (u^{\lambda+1})^2 + \cdots + (u^n)^2$. (Note that $f = c - \xi + \eta$).
We have $F \leqslant f = c - \xi + \eta \leqslant c + \frac{1}{2}\xi + \eta \leqslant c + \varepsilon$ in $\mathrm{supp}(F - f)$ (where $\xi + 2\eta \leqslant 2\varepsilon$), so $F^{-1}(-\infty, c+\varepsilon] = f^{-1}(-\infty, c+\varepsilon]$.
Step2: We show that the critical points of F are the same as those of f, hence $F^{-1}([c-\varepsilon, c+\varepsilon])$ has no critical points.
Note that $dF = \frac{\partial F}{\partial \xi} d\xi + \frac{\partial F}{\partial \eta} d\eta$ where $\frac{\partial F}{\partial \xi} = -1 - \mu'(\xi + 2\eta) < 0$ and $\frac{\partial F}{\partial \eta} = 1 - 2\mu'(\xi + 2\eta) \geqslant 1$. Hence it has no critical points in U except at p. By Step1 and $F \leqslant g$ we infer $F^{-1}([c-\varepsilon, c+\varepsilon]) \subset f^{-1}([c-\varepsilon, c+\varepsilon])$, so $F^{-1}([c-\varepsilon, c+\varepsilon])$ is compact. Furthermore, $F(p) = c - \mu(0) < c - \varepsilon$. Therefore $F^{-1}([c-\varepsilon, c+\varepsilon])$ has no critical points, so by Theorem3.1 we know $F^{-1}(-\infty, c-\varepsilon]$ is the deformation retract of $M^{c+\varepsilon} = F^{-1}((-\infty, c+\varepsilon])$.
Denote $F^{-1}(-\infty, c+\varepsilon] = M^{c-\varepsilon} \cup H$ where H is the closure of $F^{-1}(-\infty, c-\varepsilon] \setminus M^{c-\varepsilon}$.
Step3: Now consider the cell e^λ of all $q \in U$ with $\xi(q) \leqslant \varepsilon, \eta(q) = 0$. Note that $e^\lambda \subset H$ since using $\frac{\partial F}{\partial \xi} < 0$ we have $F(q) \leqslant F(p) < c - \varepsilon$ but $f(q) \geqslant c - \varepsilon$. It suffices to show $M^{c-\varepsilon} \cup e^\lambda$ is a deformation retract of $M^{c-\varepsilon} \cup H$.
Consider $r_t : M^{c-\varepsilon} \cup H \to M^{c-\varepsilon} \cup H$ which is the identity outside U, and within U, define as follows:

- Case1: If $\xi \leqslant \varepsilon$, define $r_t : (u^1, \cdots, u^n) \mapsto (u^1, \cdots, u^\lambda, tu^{\lambda+1}, \cdots, tu^n)$.
- Case2: If $\varepsilon \leqslant \xi \leqslant \eta + \varepsilon$ let $r_t : (u^1, \cdots, u^n) \mapsto (u^1, \cdots, u^\lambda, s_t u^{\lambda+1}, \cdots, s_t u^n)$ where $s_t = t + (1-t)\sqrt{\frac{\xi - \varepsilon}{\eta}} \in [0, 1]$.
- Case3: If $\xi \geqslant \eta + \varepsilon$, let $r_t = id$.

Then r_t defines a deformation retract from $M^{c-\varepsilon} \cup H$ to $M^{c-\varepsilon} \cup e^\lambda$. $\square$

Interesting observation: Taking $f \mapsto -f$, then the index $\lambda \mapsto n - \lambda$ so the number of $\lambda-$cells and $n - \lambda$ cells in the homotopy type are the same.

6

Homotopy Types from Critical Values

2026/4/24

Milnor *Morse Theory* Chapters 3.3,4

CW-complexes and homotopy groups

Def: (CW-complex)
(1) Let X^0 be a discrete set of 0-cells.
(2) There is a set of n-cells $\{(e^n_\alpha, \varphi^n_\alpha)\}_{\alpha \in I}$ where e^n_α is an n-cell and $\varphi^n_\alpha : \partial D^n \to X^{n-1}$, then $X^n = X^{n-1} \coprod_\alpha D^n_\alpha = X^{n-1} \coprod_\alpha e^n_\alpha$.
(3) $X = \bigcup_n X^n$, and $A \subset X$ is open/closed if $A \cap X^n$ is open/closed in X^n.
Remark: C - closure finiteness: if $A \subset X$ is compact, then A is contained in a finite sub-complex. W - weak topology, i.e. condition (3).

Def: (Homotopy groups) X a topological space, $x \in X$, then $\pi_n(X, x) = [(\mathbb{S}^n, *), (X, x)]$ (equivalence under homotopy). Since $(\mathbb{S}^n, *) \cong (I^n, \partial I^n)$ where $I = [0, 1]$, we can paste two cubes together alone a face. Unit: $a \in \mathbb{S}^n \mapsto x$, inverse is a reflection of cubes.
For $n \geqslant 2$, $\pi_n(X)$ is commutative.
Theorem: If X is connected ($\pi_0(X) = 1$), then $\pi_1(X)_{ab} = H_1(X)$.

Three basic theorems:
Theorem1:(Whitehead) For simply connected X, Y, $f : X \to Y$, $f_* : H_*(X) \to H_*(Y)$ is an isomorphism, then $\pi_*(X) \cong \pi_*(Y)$. ($* = i \geqslant 1$)
Theorem2: (CW approximation) If X is a topological space, there exists $\tilde{X}$ a CW-complex and $f : \tilde{X} \to X$, such that $\pi_*(\tilde{X}) \cong \pi_*(X)$.
Theorem3: (Cellular approximation) If X, Y are CW-complexes, $f : X \to Y$, then exists $f \cong g$ under homotopy, such that $g : X \to Y$ is cellular, i.e., $g(X^n) \subset Y^n$.

Homotopy types

Theorem3.3: If $f : M \to \mathbb{R}$ without degenerate critical points, and each M^a is compact, then M is homotopy equivalent to a CW complex with one $n-$cell of a critical point of index n.

Lemma1: $\varphi_0, \varphi_1 : \partial e^\lambda \to X$, $\varphi_0 \cong \varphi_1$, then $X \cup_{\varphi_0} e^\lambda \cong X \cup_{\varphi_1} e^\lambda$.
Lemma2: $\varphi : \partial e^\lambda \to X$, $f : X \cong Y$, then f extends to $F : X \cup_\varphi e^\lambda \to Y \cup_{f \circ \varphi} e^\lambda$.
See textbook for proof.

Proof of Theorem3.3: Let $c_1 < c_2 < \cdots$ be critical values of $f : M \to R$, and $M_i = M^{c_i + \varepsilon_i}$.

By Theorem3.1, $M^{c_i+\varepsilon_i} \cong M^{c_{i+1}-\varepsilon_{i+1}}$, and by Theorem3.2, $M^{c_i+\varepsilon_i} \cong M^{c_i-\varepsilon_i} \cup_{\varphi_1} e^{\lambda_1} \cup \cdots \cup_{\varphi_{j(c)}} e^{\lambda_{j(c)}}$. By Lemma2, $M^{c_i-\varepsilon_i} \cup_{\varphi_i} e^{\lambda_i} \cong M^{c_{i-1}+\varepsilon_{i-1}} \cup_{g_i \circ \varphi_i} e^{\lambda_i} \cong M_{i-1} \cup_{\varphi'} e^{\lambda_i}$ where φ' is the cellular approximation.
Let $X_i = M_i$, we obtain a CW complex X. And by induction it follows that M has homotopy type X. $\square$
Counterexamples when non-compact: (a) $f : \mathbb{S}^1 \times \mathbb{R} \to \mathbb{R}, (x,t) \mapsto t$ then f has no critical values but $\mathbb{S}^1 \times \mathbb{R}$ is non-trivial, failure caused by an initial non-triviality; (b) $f : \mathbb{R}^2 \backslash \{1,1\} \to \mathbb{R}, (x,y) \mapsto x^2+y^2$, then f has one critical point of index zero, but $\mathbb{R}^2 \setminus \{(1,1)\} \simeq \mathbb{S}^1$, failure caused by non-compactness of M^2.

Some Examples

Theorem4.1: (Reeb) If M is a compact manifold and $f : M \to \mathbb{R}$ has 2 critical points, both are non-degenerate, then M is homeomorphic to a sphere $\mathbb{S}^n$. (though maybe not with the standard differential structure of $\mathbb{S}^n$, i.e. the exotic sphere $\mathbb{S}^7$)
Proof: Clearly the two critical points can only take minimal and maximal values, say $f(p) = 0$ and $f(q) = 1$. For small ε, $f^{-1}([0,\varepsilon])$ and $f^{-1}([1-\varepsilon,1])$ are closed 0-cell and $n-$cells by Theorem3.1.

Analogously, if there are only three critical points, then by duality they have index $0, n, \frac{n}{2}$, so it has homotopy type of an $\frac{n}{2}-$sphere with an $n-$cell attached.

Now we consider the homotopy type of $\mathbb{CP}^n$: View it as $\left\{(z_0 : \cdots : z_n) : \sum |z_j|^2 = 1\right\}$. Consider $f(z_0 : \cdots : z_n) = \sum c_j |z_j|^2$, where c_i are distinct real numbers.
Consider the local coordinate $z_0 \neq 0$ at $p_0 = (1 : 0 : \cdots : 0)$, set $\frac{z_j}{z_0}|z_0| = x_j + iy_j$, then $x_1, y_1, \cdots, x_n, y_n$ are the coordinates, and $f = c_0 + \sum_{j=1}^n (c_j - c_0)(x_j^2 + y_j^2)$. Since c_i are distinct, the only critical point in this coordinate is p_0. Analogously, the critical points of f are p_i and the index is twice the number of j with $c_j < c_i$. By Theorem3.3, $\mathbb{CP}^n$ has homotopy type $e^0 \cup e^2 \cup \cdots \cup e^{2n}$, hence has homology group $H_i(\mathbb{CP}^n;\mathbb{Z}) = \mathbb{Z}$ for $i = 0, 2, \cdots, 2n$ and otherwise $H_i(\mathbb{CP}^n;\mathbb{Z}) = 0$.

7

Existence of Morse Functions

2026/4/30

Existence of Morse Function

Goal: For any smooth manifold there is a Morse function, i.e., every critical point is non-degenerate.

Theorem: Given a smooth Whitney embedding $f : M \to \mathbb{R}^{n+k}$ where M is a n-dim manifold, then for almost any point $p \in \mathbb{R}^{n+k}$, the distance function $L_p : M \to \mathbb{R}, x \mapsto \|x - p\|^2$ is a Morse function.

First do some calculations: Let the local coordinates of M be $(u_1, \cdots, u_n)$ and embedded into $(x_1, \cdots, x_{n+k})$, then $\frac{\partial}{\partial u_i} L_p = 2 \frac{\partial x}{\partial u_i} \cdot (x - p)$, and $\frac{\partial^2}{\partial u_i \partial u_j} L_p = 2 \left(\frac{\partial^2 x}{\partial u_i \partial u_j} \cdot (x - p) + \frac{\partial x}{\partial u_i} \cdot \frac{\partial x}{\partial u_j} \right)$. On a critical point, this is just $g_{ij} = \frac{\partial x}{\partial u_i} \cdot \frac{\partial x}{\partial u_j}$ and $l_{ij}(v) = \frac{\partial^2 x}{\partial u_i \partial u_j} \cdot v$. Hence the Hessian of L_p at x is $g - tl(v)$ if we denote $x - p = t\vec{v}$.
We want to the points such that $g - tl(v)$ is degenerate. It seems like centers of curvature spheres of curves on the manifold.

Proof of Theorem: Consider $E : N \to \mathbb{R}^{n+k}$ where N is the normal bundle of $f : M \to \mathbb{R}^{n+k}$, defined by $E : (x, v) \mapsto x + v$. We show that the critical values of E are those we want to expel. Then it is trivial by Sard's theorem.
Let the local coordinates of N be $(u_1, \cdots, u_n, t_1, \cdots, t_k) \mapsto x + t_1\vec{w}_1 + t_2\vec{w}_2 + \cdots + t_k\vec{w}_k$ where $\vec{w}_i : (u_1, \cdots, u_n) \to \mathbb{R}^{n+k}$ such that $\|\vec{w}_i(q)\|^2 = 1$ and $\vec{w}_i(q) \cdot \vec{w}_j(q) = 0 \forall i \neq j$. Then

$$\frac{\partial E}{\partial u_i} = \frac{\partial x}{\partial u_i} + \sum_{j=1}^{k} t_j \frac{\partial w_j}{\partial u_i}, \frac{\partial E}{\partial t_j} = w_j.$$

Take inner product with bases $\left(\frac{\partial x}{\partial u_1}, \cdots, \frac{\partial x}{\partial u_n}, w_1, \cdots, w_k \right)$, then the matrix becomes

$$\begin{pmatrix} \frac{\partial x}{\partial u_i} \cdot \frac{\partial x}{\partial u_j} + \sum_{\alpha=1}^{k} t_\alpha \frac{\partial w_\alpha}{\partial u_i} \cdot \frac{\partial x}{\partial u_j} & \cdots \\ 0 & id \end{pmatrix}$$

and its rank does not change. Note that $0 = \frac{\partial}{\partial u_i} \left(w_\alpha \cdot \frac{\partial x}{\partial u_j} \right)$ implies $\frac{\partial w_\alpha}{\partial u_i} \cdot \frac{\partial x}{\partial u_j} = -w_\alpha \cdot \frac{\partial^2 x}{\partial u_i \partial u_j}$. Hence the formula becomes $g_{ij} - (\sum t_\alpha \vec{w}_\alpha) \cdot \frac{\partial^2 x}{\partial u_i \partial u_j}$. Hence the critical values are just the image points where L_p has a degenerate critical point.

8

The Poincaré–Hopf Theorem

Reference: Milnor, Topology from a Differential Viewpoint & Morse Theory.

Index of a Vector Field

Recall the theorem of Sard and Brown:
Theorem (Sard): If $f : M \to \mathbb{R}^n$ is smooth, then the set of critical values has Lebesgue measure 0.
Corollary: If $f : M \to N$ is smooth, then the set of regular values is dense in N.

We can define the degree of a smooth map.
Theorem: If M, N are $n-$dimensional compact connected oriented smooth manifolds, and $F : M \to N$ is smooth. For $x \in M$ let $\text{sign}(x) = +1$ if dF_x is orientation preserving and $\text{sign}(x) = -1$ otherwise. Then for any regular value $y \in N$,

$$\deg(F) := \sum_{x \in F^{-1}(y)} \text{sign}(x)$$

is independent of the choice of y. Furthermore, if F, G are smoothly homotopic, then $\deg(F) = \deg(G)$.
Remark: The degree is actually defined by

$$\int_M F^*\omega = \deg(F) \int_N \omega$$

for any smooth $n-$form ω on N. Then taking ω as a compact cover of $F^{-1}(y)$ we obtain the result above. See GTM218 page 458.

Lemma1: If $f : M \to N$ extends to $F : X \to N$, where $M = \partial X$ is the boundary of a compact smooth manifold X with the same orientation, then $\deg(f) = 0$.
Proof: Take a $n-$form such that $\int_N \omega \neq 0$, then $f = F \circ \iota$ implies

$$\deg(f) \int_N \omega = \int_M f^*\omega = \int_{\partial X} F^*\omega = \int_X d(F^*\omega) = \int_X F^* d\omega = 0.$$

(The third equality is Stokes' Theorem) Since the only $(n+1)-$form on the n-dim space N is zero. Hence $\deg(f) = 0$.

Now we define the index of a vector field at its zero:
Definition: If $U \subset \mathbb{R}^n$ is open, v a vector field on U, and z an isolated zero. Take $\varepsilon > 0$ such that

$B(z,\varepsilon)$ contains no other zeros, then the index $\mathrm{ind}_v(z)$ of v at z is defined as the degree of the map

$$\varphi : \partial B(x,\varepsilon) \to \mathbb{S}^{n-1}, y \mapsto \frac{v(y)}{\|v(y)\|}.$$

Examples: Consider electromagnetic fields.

Lemma2: Suppose $U, V \subset \mathbb{R}^n$ open, $f : U \to V$ a diffeomorphism, v a vector field on U. Let $v' = df \circ v \circ f^{-1}$ be a vector field on V, then $\mathrm{ind}_{v'}(f(z)) = \mathrm{ind}_v(z)$.
Lemma3: Any orientation preserving diffeomorphism of $\mathbb{R}^m$ is smoothly isotopic to the identity.
Proof of Lemma3: For any $f : \mathbb{R}^m \to \mathbb{R}^m$ WLOG assume $f(0) = 0$. Consider $F(x,t) = f(xt)/t$ for $0 < t \leqslant 1$ and $F(x,0) = df_0(x)$, then since $df_0(x) = \lim_{t\to 0} f(xt)/t$, F is clearly an isotopy. To show that F is smooth as $t \to 0$, write f as $f(x) = x_1 g_1(x) + \cdots + x_n g_n(x)$ where $g_k(x) = \int_0^1 \frac{\partial f}{\partial x_k}(tx)\,\mathrm{d}t$, then $F(x,t) = x_1 g_1(tx) + \cdots + x_n g_n(tx)$ for all $t \in [0,1]$. Hence f is isotopic to the linear map df_0, so it is isotopic to the identity.
Proof of Lemma2: We may assume $z = f(z) = 0$ and U is convex.
First assume f is orientation preserving. By Lemma3 we have embeddings $f_t : U \to \mathbb{R}^n$ such that $f_0 = id$, $f_t(0) = 0$ and $f_1 = f$. Let $v_t = df_t \circ v \circ f_t^{-1}$ be vector fields, then there exists $\varepsilon > 0$ such that the only zero of each v_t in $B(0,\varepsilon) \subset \mathrm{Im} f_t$ is $z = 0$. (Such ε exists: Consider a closed ball $D \subset U$ centered at 0, then $f_t(\partial D)$ is jointly continuous so the image of $\partial D \times [0,1]$ is compact, hence the distance from 0 to the image is positive. Therefore we can take $\varepsilon > 0$ such that $B(0,\varepsilon) \subset f_t(D)$.)
Then since homotopic maps have the same degree, the Gauss maps of v_0 and v_1 have the same degree, hence $\mathrm{ind}_{v'}(f(z)) = \mathrm{ind}_v(z)$.
To prove the orientation reversing case, it suffices to consider when $f = \rho$ is a reflection. Then the associated gauss map $\bar{v}' = \rho \circ \bar{v} \circ \rho^{-1}$ so we can verify they have the same degree.

By this lemma, the index of vector fields on manifolds is well-defined:
Definition: If z is an isolated zero of v on M, and (U,φ) a local coordinate, then $\mathrm{ind}_v(z)$ is defined as the index of $d\varphi \circ v \circ \varphi^{-1}$ at $\varphi(z)$.
For a vector field v, denote its index as the sum of indices of zeros of v. (as a shorthand)

Definition: If $M \subset \mathbb{R}^m$ is a compact $m-$dimensional manifold with boundary, then the Gauss map is defined as $g : \partial M \to \mathbb{S}^{m-1}$ which maps x to the normal unit vector, the outward vector perpendicular to $T(\partial M_x)$
Theorem: Let $M \subset \mathbb{R}^m$ be a $m-$dimensional manifold with boundary, v a vector field that points outward on the boundary, with isolated zeros. Then the index of v is the degree of the Gauss map.
Proof: Use Hausdorffness to find small spheres containing a unique zero of v. By removing these zeros we obtain another manifold N with boundary. Consider $F : N \to \mathbb{S}^{m-1}, x \mapsto v(x)/\|v(x)\|$, then F extends from $F\big|_{\partial N}$ so by Lemma1 $\deg(F\big|_{\partial N}) = 0$.
Since v points outward on ∂M, $F\big|_{\partial M}$ is homotopic to the Gauss map g: e.g. consider $W : \partial M \times [0,1] \to \mathbb{R}^m, (x,t) \mapsto (1-t)\frac{v(x)}{\|v(x)\|} + tg(x)$ and the homotopy $H(x,t) = W(x,t)/\|W(x,t)\|$. Hence $\deg(F\big|_{\partial M}) = \deg(g)$, and by definition $\deg(F\big|_{C(z,\varepsilon)}) = -\mathrm{ind}_v(z)$ for any zero z of v (since the orientation reversed when looking from the outside). Therefore $\deg(F\big|_{\partial N}) = 0$ implies $\sum \mathrm{ind}_v(z) = \deg(g)$.

Corollary: Let $M \subset \mathbb{R}^k$ be a $m-$dimensional manifold without boundary, v a vector field on M with no degenerate zeros. For small $\varepsilon > 0$ define $N_\varepsilon = \{x \in \mathbb{R}^k : \|x - y\| \leqslant \varepsilon \text{ for some } y \in M\}$, then the index of v is the degree of the Gauss map on N_ε.
Proof: Consider the projection $\pi : N_\varepsilon \to M$ sending $x \in N_\varepsilon$ to its unique nearest neighbor, then π is smooth and $x - \pi(x)$ is orthogonal to $T_{\pi(x)}M$. Let $V(x) = v(\pi(x)) + (x - \pi(x))$ be a vector field on N_ε, then by orthogonality we know the zeros of V must be zeros of v.
Furthermore, suppose $v(z) = 0$, if $x \in T_z M$ then $dV(x) = dv(x)$ and if $x \in T_z M^\perp$ then $dV(x) = x$. Hence the index of V and v at z are the same.
Verify that V points outwards on ∂N_ε: $n(x) = \frac{x-\pi(x)}{\varepsilon}$ so $V(x) \cdot n(x) = (v(\pi(x)) + (x - \pi(x))) \cdot n(x) =$

$\frac{(x-\pi(x))^2}{\varepsilon} = \varepsilon > 0$.

Since the Gauss map is independent of the vector field, we see that the index of vector field with no degenerate zeros are the same.

Applying Morse theory

For any Morse function f, denote $\mathfrak{i}_{f,j}$ as the number of critical points with index j.
Theorem: For any closed manifold M and Morse function $f : M \to \mathbb{R}$,

$$\sum_j (-1)^j \mathfrak{i}_{f,j} = \chi(M).$$

Proof: By Morse theory, $\mathfrak{i}_{f,j}$ is the number of $j-$cells attached when constructing the CW-complex M, so by definition it is the Euler characteristic.

As a corollary, we calculate the Euler characteristic for some manifolds:

- If M is a closed manifold of odd dimension, then $\chi(M) = 0$, since $-f$ is a Morse function, and $\mathfrak{i}_{f,j} = \mathfrak{i}_{-f,n-j}$.
- If $M = \mathbb{S}^{2n}$, then consider the height function f, with two critical points at poles. Hence $\chi(M) = (-1)^0 + (-1)^{2n} = 2$.
- If $M = \mathbb{T}$, likewise choose the height function f, then $\chi(\mathbb{T}) = (-1)^0 + (-1)^1 + (-1)^1 + (-1)^2 = 0$.

Theorem: Let M be a compact smooth manifold, there is a vector field v such that the index of v is $\chi(M)$.
Proof: Take a Morse function $f : M \to \mathbb{R}$, and let $v = \operatorname{grad} f$. Then for each zero x of v, suppose the Morse theory index of x is j, then $\operatorname{ind}_v(x) = (-1)^j$ since the Morse function is locally of the form $f(x) \pm x_1^2 \pm x_2^2 \pm \cdots$. Therefore $\sum \operatorname{ind}_v(x) = \sum_j (-1)^j \mathfrak{i}_{f,j} = \chi(M)$.

Poincaré-Hopf Theorem

Theorem (Poincaré-Hopf) Let v be a vector field on a compact oriented smooth manifold M (without boundary), having only isolated zeros. Then the sum of indices of the zeros of the vector field is equal to $\chi(M)$.
Proof: If all zeros of v are non-degenerate, by Morse theory we found v' such that the sum of indices of zeros of v' is $\chi(M)$, and we showed that the sum of indices of zeros of v' and v are the same. So this case was already proved.
Now suppose v has a degenerate zero at $z \in M$, and take a local coordinate $U \subset \mathbb{R}^k$ at z. Since the zeros are isolated, take $\varepsilon > 0$ such that $\overline{B}(z, 2\varepsilon) \subset U$ contains no other zero of v. Take $\delta > 0$ such that $\|v(x)\| > \delta \forall x \in \overline{B}(x, 2\varepsilon) \backslash B(x, \varepsilon)$. By Urysohn's lemma, we can take $f : U \to [0, 1]$ such that $f|_{\overline{B}(x,\varepsilon)} \equiv 1$ and $f|_{B(x,2\varepsilon)^C} \equiv 0$. Consider the vector field $v'(x) = v(x) - f(x)y$ for $x \in U$ and $v' = v$ outside U, where y is a regular value of v such that $\|y\| < \delta$ (by Sard's theorem such y exists).
We show that v' has no degenerate zeros in U: If $x \in U \backslash \overline{B}(x, 2\varepsilon)$, then $v'(x) = v(x) \neq 0$. If $x \in \overline{B}(x, 2\varepsilon) \backslash B(x, \varepsilon)$, then $\|v(x)\| > \delta > \|y\|$ so $v'(x) = v(x) - f(x)y \neq 0$. If $x \in B(x, \varepsilon)$, then $v(x) = y$ so x is a regular point of v, hence $dv'_x = dv_x \neq 0$ so x is non-degenerate.
It suffices to show the sum of indices of zeros of z, z' coincide, then by repeatedly removing a degenerate zero, we obtain a vector field without degenerate zeros. The proof is analogous to the prior theorem.
By definition $\operatorname{ind}_v(z)$ is the degree of $F : \partial\overline{B}(z, 2\varepsilon) \to \mathbb{S}^{n-1}, x \mapsto \frac{v(x)}{\|v(x)\|}$. Again we take small

spheres in each zero z' of v', and remove them to obtain a smooth manifold N with boundary. Likewise $\deg(F\big|_{\partial N}) = 0$ implies $\deg(F\big|_{\partial \overline{B}(0,2\varepsilon)}) = \sum \mathrm{ind}_{v'}(z')$ so $\mathrm{ind}_v(z) = \sum \mathrm{ind}_{v'}(z')$.

We have the following famous result as a corollary:
Theorem (Hairy Ball): Any smooth vector field on $\mathbb{S}^{2n}$ has at least 1 zero.
Proof: By Poincaré-Hopf the sum of indices of zeros is $\chi(\mathbb{S}^{2n}) = 2$, so the vector field cannot be non-zero.
We can easily construct a non-zero vector field on $\mathbb{S}^{2n-1} \subset \mathbb{R}^{2n}$: For $x = (x_1, \cdots, x_{2n})$, let $v(x) = (-x_2, x_1, -x_4, x_3, \cdots, -x_{2n}, x_{2n-1})$, then $v(x) \cdot x = 0$ and $\|v(x)\| = \|x\| = 1$.

9

Introduction to Riemannian Geometry

2026/5/7

Def: A **Riemannian metric** on a smooth manifold M is a correspondence $p \in M \mapsto \langle \cdot, \cdot \rangle_p$ which is an inner product on T_pM, such that in local coordinates $(x_1, \cdots, x_n)$ at q, $\left\langle \frac{\partial}{\partial x_i}(q), \frac{\partial}{\partial x_j}(q) \right\rangle_q = g_{ij}(x_1, \cdots, x_n)$ is a smooth function on U.
An isometry is a diffeomorphism $f : M \to N$ that keeps the metric, i.e. $\langle u, v \rangle = \langle df_p(u), df_p(v) \rangle_{f(p)}$ for all $p \in M, u, v \in T_pM$.

Def: A vector field along a curve $c : I \to M$ is a vector field on c, and its velocity field is $\frac{dc}{dt} = dc\left(\frac{d}{dt}\right)$.

Prop: Any smooth manifold has a Riemannian metric.
Proof: Take a partition of unity $\{(V_\alpha, f_\alpha)\}$, then clearly we can define a Riemannian metric $\langle \cdot, \cdot \rangle^\alpha$ on each V_α, the metric induced by local coordinates. Let $\langle u, v \rangle_p = \sum_\alpha f_\alpha(p) \langle u, v \rangle_p^\alpha$ for all $p \in M, u, v \in T_pM$, verify that it defines a Riemannian metric on M.

Let $\mathfrak{X}(M)$ be all smooth vector fields on M, and $D(M)$ the ring of smooth $f : M \to \mathbb{R}$.
Def: An **affine connection** ∇ on a smooth manifold M is a mapping $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ denoted by $(X, Y) \mapsto \nabla_X Y$ such that

- (i) $\nabla_{fX+gY} Z = f \nabla_X Z + g \nabla_Y Z$;
- (ii) $\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z$;
- (iii) $\nabla_X (fY) = f \nabla_X Y + X(f) Y$.

Prop: There exists a unique correspondence: a vector field V along the curve $c : I \to M$ assigned to another vector field $\frac{DV}{dt}$ along c, called the covariant derivative of V along c, such that

- (a) $\frac{D}{dt}(V + W) = \frac{DV}{dt} + \frac{DW}{dt}$;
- (b) $\frac{D}{dt}(fV) = \frac{df}{dt} V + f \frac{DV}{dt}$;
- (c) If $V = Y \circ c$ for some $Y \in \mathfrak{X}(M)$, then $\frac{DV}{dt} = \nabla_{\frac{dc}{dt}} Y$.

If V is along $c : I \to M$ and $\frac{DV}{dt} \equiv 0$ we call it **parallel** to c.

- For $V_0 \in T_{c(t_0)}M$, there is a unique parallel V along c such that $V(t_0) = V_0$. $V(t)$ is called a parallel transport of $V(t_0)$ along c.

If, for any c and any p, p' parallel along c, $\langle p, p' \rangle$ is constant, then we call $M, \nabla, \langle \cdot, \cdot \rangle$ **compatible**. Or equivalently, for any V, W along c, we have the Leibniz rule $\frac{d}{dt}\langle V, W \rangle = \left\langle \frac{DV}{dt}, W \right\rangle + \left\langle V, \frac{DW}{dt} \right\rangle$.

Call an affine connection ∇ **symmetric** if $\nabla_X Y - \nabla_Y X - [X,Y] = 0 \forall X, Y$.
Theorem: (Levi-Civita) For any Riemannian manifold M, there is a unique connection ∇ on M such that it is symmetric and compatible.

Def: A curve $\gamma : I \to M$ is a geodesic at $t_0 \in I$, if $\frac{D}{dt}\left(\frac{d\gamma}{dt}\right) = 0$. If γ is a geodesic for all $t_0 \in I$, we call it a geodesic connecting $\gamma(a), \gamma(b)$.
By ODE theory, there is a unique geodesic field G on TM such that on every point G is of the form $t \mapsto (\gamma(t), \gamma'(t))$ where γ is a geodesic on M. Its flow is called a geodesic flow.

10

Basic Riemannian Geometry

2026/5/8

Part1 Riemannian Connection

Notation: The Riemannian metric is $g = g_{ij} dx^i \otimes dx^j$ such that $g_{ij} = g_{ji}$. Any vector field can be written as $X = \sum X^i \frac{\partial}{\partial x^i}$.
A Riemann connection must satisfy:
1. $\nabla_X fY = X(f) + f \nabla_X Y$;
2. $\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z$;
3. $\nabla_{X+fY} Z = \nabla_X Z + f \nabla_Y Z$;
Define $\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \sum_k \Gamma_{ji}^k \frac{\partial}{\partial x^k}$ as the Christoffel notation. Then (1)(2)(3) can be written as $\nabla_X Y = X^i \left(\frac{\partial Y^j}{\partial x^i} \frac{\partial}{\partial x^j} + Y^j \Gamma_{ji}^k \frac{\partial}{\partial x^k} \right)$.
4. $\nabla_X Y - \nabla_Y X = [X, Y]$; which means $\Gamma_{ji}^k = \Gamma_{ij}^k$;
5. $\nabla_X \langle Y, Z \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle$; which means $\nabla_{\frac{\partial}{\partial x^k}} \left\langle \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle = \frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ki}^l g_{lj} + \Gamma_{kj}^l g_{il}$, i.e. $\Gamma_{ij}^k = \frac{1}{2} g^{kl} \left(\frac{\partial g_{li}}{\partial x^j} + \frac{\partial g_{lj}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right)$.

Part2 Minimal is Geodesic

Consider $\gamma : I \to M$, we can define $L(\gamma) = \int_0^1 \|\gamma'\| \, \mathrm{d}t$.
For any smooth map $\Phi : I \times [0, \varepsilon) \to M, (t, s) \mapsto \Phi(s, t)$ such that $\Phi(\cdot, 0) = \gamma$, let $T = \frac{\partial \Phi}{\partial t}, S = \frac{\partial \Phi}{\partial s}$.
Consider

$$\frac{d}{ds}|_{s=0} L = \int_0^1 \frac{d}{ds} \sqrt{\langle T, T \rangle} \, \mathrm{d}t = \int_0^1 \frac{1}{\|T\|} \langle \nabla_S T, T \rangle \, \mathrm{d}t = \int_0^1 \frac{1}{\|T\|} \langle \nabla_T S, T \rangle \, \mathrm{d}t \tag{10.1}$$

$$= \frac{1}{\|T\|} \langle S, T \rangle \Big|_0^1 - \frac{1}{\|T\|} \int_0^1 \langle S, \nabla_T T \rangle \, \mathrm{d}t. \tag{10.2}$$

Hence if γ is locally minimal and of constant speed, the derivative is zero, so $\nabla_T T = 0$ which implies γ is a geodesic.

Part3 Exp map and Gauss lemma

For $p \in M$, any $v \in T_pM$ determines $\gamma : I \to M$ such that $\gamma(0) = p, \gamma'(0) = v$ and $\nabla_{\dot\gamma}\dot\gamma = 0$. Hence we can define $\exp_p : O \subset T_pM \to M, v \mapsto \gamma(1; p, v)$. By ODE theory, exp is a smooth map. $\exp_p : B_0(\delta) \to M$ is a local diffeomorphism for small δ.
Lemma (Gauss lemma): $\langle \exp_{p*}\big|_v v, \exp_{p*}\big|_v w\rangle = \langle v, w\rangle$.
Proof: Consider $\Phi = \exp_p(t(v+sw))$ then $T = \exp_{p*}\big|_{tv} v$, $S = \exp_{p*}\big|_{tv} tw$, so $\frac{d}{ds}L = \frac{1}{\|v\|}\langle S, T\rangle\big|_0^1 = \frac{1}{\|v\|}\langle \exp\cdots, \exp\cdots\rangle$. Meanwhile, by definition $\frac{d}{ds}L = \int_0^1 \frac{d}{ds}\|v+sw\|\,\mathrm{d}t = \frac{1}{\|v\|}\langle v, w\rangle$.

Part4 Hopf-Rinow Theorem

For $p, q \in M$ define $d(p,q) = \inf_{\gamma(0)=p,\gamma(1)=q} L(\gamma)$ to be their distance. For $q = \exp_p v$, we want $d(p,q) = L(\gamma)$ where γ is the geodesic.
Theorem: For (M, g) Riemannian manifold, $p \in M$, then there exists δ such that $\exp_p : B_0(\delta) \subset T_pM \to M$ is a smooth embedding s.t. $\exp(B_0(\delta)) = B_p(\delta) = \{q : d(p,q) < \delta\}$, and for any $q \in B_p(\delta)$ there exists γ geodesic such that $q = \exp_p v, \gamma = \exp_p tv$ and $d(p,q) = L(\gamma)$.
Theorem: There is a geodesically convex neighborhood, i.e., some $\delta \ll 1$ such that $B_p(\delta)$ is convex, which means for any two points and any minimal curve connecting $q_1, q_2 \in B_p(\delta)$, the curve lies in $B_p(\delta)$.

Theorem: (Hopf-Rinow) For (M, g) TFAE: (1) (M, d) is complete as a metric space; (2) $\forall p \in M$, $\exp_p : T_pM \to M$ is well-defined on the whole T_pM; (3) exists $p \in M$, $\exp_p : T_pM \to M$ is well-defined; (4) every bounded and closed set in M is compact.

11

Energy Variations

2026/5/15

Morse Theory Chapters 11-13

Def: Curvature: The curvature R of a Riemannian manifold (M, g) is a correspondence associating each pair $(X, Y) \in \mathfrak{X}(M)$ a mapping $R(X, Y) : \mathfrak{X}(M) \to \mathfrak{X}(M), Z \mapsto D_X D_Y Z - D_Y D_X Z - D_{[X,Y]} Z$.

Def: Sectional Curvature: Let $\sigma \subset T_p M$ be a two-dimensional subspace of $T_p M$ and $x, y \in \sigma$ independent. Then we define $K(x, y) = \langle R(x, y)x, y\rangle / (|x|^2 |y|^2 - \langle x, y\rangle^2)$. It is sort of like $\langle R(x, y)x, y\rangle$ represents the area of some region of M, and $|x|^2|y|^2 - \langle x, y\rangle^2$ is the area of x, y in $\mathbb{R}^2$.
Fact: For a surface, K is the Gauss curvature. Let M^n be a complete Riemannian manifold of even dimension with constant $K = 1$. Then M^n is isometric to $\mathbb{S}^n$ or $\mathbb{RP}^n$.

Path Space and Energy

For a smooth manifold M, and $p, q \in M$, let $\Omega(M; p, q)$ be all piece-wise smooth paths from p to q. Later we will give a topology on Ω, for now we define the **tangent space** of Ω at a path ω as $T\Omega_\omega = \{W \text{ vector field along } \omega : W(0) = W(1) = 0\}$.

Recall: For a piecewise smooth path $\omega : [0, 1] \to M$, a variation of ω is a function $\alpha : (-\varepsilon, \varepsilon) \times [0, 1] \to M$ for some $\varepsilon > 0$ such that (1) $\alpha(0, \cdot) = \omega$; (2) there is a subdivision $0 = t_0 < \cdots < t_k = 1$ of $[0, 1]$ such that the map α is C^∞ on each $(-\varepsilon, \varepsilon) \times [t_i, t_{i+1}]$; (3) $\alpha(u, 0) = p, \alpha(u, 1) = q \forall u \in (-\varepsilon, \varepsilon)$.
$W_t = \frac{\partial \alpha}{\partial u}(0, t)$ is called a **variation vector field**.

Proposition: Assume (M, g) is a Riemannian manifold, $\omega : [0, 1] \to M$ piecewise smooth, $U = U(t)$ is a piecewise smooth vector field along ω, $U(t) \in T_{\omega(t)} M$, then there exists a variation $\alpha : (-\varepsilon, \varepsilon) \times [0, 1] \to M$ such that $U(t)$ is its variation vector field.
Proof: Consider the exponential map, define $\alpha : (-\varepsilon, \varepsilon) \times [0, 1] \to M, (u, t) \mapsto \exp_{\omega(t)}(uU(t))$ and verify it is piecewise smooth.

Then for $F : \Omega \to \mathbb{R}$ we can define $F_* : T\Omega_\omega \to T\mathbb{R}_{F(\omega)}$ as follows: Given $W \in T\Omega_\omega$ choose a variation $\alpha : (-\varepsilon, \varepsilon) \to \Omega$ with $\alpha(0, \cdot) = w$, $\frac{d\alpha}{du}(0) = W$, and set $F_*(W)$ be $\frac{(F(\alpha(u, \cdot)))}{du}|_{u=0}$ multiplied by the tangent vector $\frac{d}{dt}_{F(\omega)}$.

Def: Length and energy: For a path $\omega : [0, 1] \to M, 0 \leqslant a < b \leqslant 1$,

$$L_a^b(\omega) = \int_a^b \|\frac{\partial \omega}{\partial t}\| \,\mathrm{d}t, E_a^b(\omega) = \int_a^b \|\frac{\partial \omega}{\partial t}\|^2 \,\mathrm{d}t.$$

Denote $V_t = \frac{\partial\omega}{\partial t}$ called the **velocity vector** and $A_t = \frac{\partial^2}{\partial t^2}\omega$ the **acceleration vector**.

Theorem (First variation formula): $\alpha : (-\varepsilon, \varepsilon) \times [0,1] \to M$ is a variation of ω. Then

$$\frac{1}{2}\frac{dE(\alpha(u,-))}{du}\Big|_{u=0} = -\sum_t \langle W_t, \Delta_t V\rangle - \int_0^1 \left\langle W_t, \frac{\partial^2}{\partial t^2}\omega \right\rangle \mathrm{d}t \qquad (\star)$$

where $\Delta_t V = V_{t+0} - V_{t-0}$.
Proof:

$$\frac{1}{2}\frac{dE(\alpha(u,-))}{du} = \frac{1}{2}\frac{d}{du}\int_0^1 \left\langle \frac{\partial\alpha}{\partial t}, \frac{\partial\alpha}{\partial t}\right\rangle \mathrm{d}t = \int_0^1 \left\langle \frac{\partial^2\alpha}{\partial u\partial t}, \frac{\partial\alpha}{\partial t}\right\rangle \mathrm{d}t$$

Suppose α is smooth on $(-\varepsilon,\varepsilon)\times[t_i, t_{i+1}]$ then (by Levi-Civita $\frac{\partial^2\alpha}{\partial u\partial t} = \frac{\partial^2\alpha}{\partial t\partial u}$)

$$\frac{\partial}{\partial t}\left\langle \frac{\partial\alpha}{\partial u}, \frac{\partial\alpha}{\partial t}\right\rangle = \left\langle \frac{\partial^2\alpha}{\partial u\partial t}, \frac{\partial\alpha}{\partial t}\right\rangle + \left\langle \frac{\partial\alpha}{\partial t}, \frac{\partial^2\alpha}{\partial t^2}\right\rangle.$$

Hence

$$\left\langle \frac{\partial\alpha}{\partial u}, \frac{\partial\alpha}{\partial t}\right\rangle\Big|_{t_{i-1}}^{t_i} = \int_{t_{i-1}}^{t_i} \left\langle \frac{\partial^2\alpha}{\partial u\partial t}, \frac{\partial\alpha}{\partial t}\right\rangle \mathrm{d}t + \int_{t_{i-1}}^{t_i} \left\langle W_t, \frac{\partial^2\alpha}{\partial t^2}\right\rangle \mathrm{d}t.$$

Summing up we obtain $(\star)$.
Corollary: The path ω is a geodesic iff it is a critical point of E.

Hessian of Energy Function

Motivation: Recall the Hessian of $f : M \to \mathbb{R}$ at its critical point is defined as $f_{**} : TM_p \times TM_p \to \mathbb{R}$: Given $X_1, X_2 \in TM_p$ choose a smooth map $(u_1, u_2) \mapsto \alpha(u_1, u_2)$ defined on a neighborhood of $(0,0)$ in $\mathbb{R}^2$, with values in M, so that $\alpha(0,0) = p$, $\frac{\partial\alpha}{\partial u_1}(0,0) = X_1$, $\frac{\partial\alpha}{\partial u_2}(0,0) = X_2$. Then $f_{**}(X_1, X_2) = \frac{\partial^2 f(\alpha(u_1,u_2))}{\partial u_1\partial u_2}\Big|_{(0,0)}$.
Now we can define E_{**} as follows: For $\omega : [0,1] \to M$. Given two variation vector fields W_1, W_2 along ω, choose a 2-parameter variation $\alpha : U \times [0,1] \to M$, where U is a neighborhood of $(0,0)$ in $\mathbb{R}^2$ so that $\alpha(0,0,t) = \omega(t)$, $\frac{\partial\alpha}{\partial u_1}(0,0,t) = W_1(t)$ and $\frac{\partial\alpha}{\partial u_2}(0,0,t) = W_2(t)$. Then the Hessian $E_{**}(W_1, W_2)$ can be defined as

$$E_{**}(W_1, W_2) = \frac{\partial^2 E(\alpha(u_1, u_2, -))}{\partial u_1 \partial u_2}\Big|_{(0,0)}$$

Fact: $E_{**}(W_1, W_2) = E_{**}(E_2, E_1)$ since we can exchange ∂u_1 and ∂u_2.

Theorem: Let $u : U \times [0,1] \to M$ be a 2-parameter variation of the geodesic ω with variation vector field $W_i(t) = \frac{\partial\alpha}{\partial u_i}(0,0,t)$ then the second derivative

$$\frac{1}{2}\frac{\partial^2 E}{\partial u_1\partial u_2}(0,0) = -\sum_t \left\langle W_2(t), \Delta_t \frac{\partial W_1}{\partial t}\right\rangle - \int_0^1 \left\langle W_2, \frac{\partial^2 W_1}{\partial t^2} + R\left(W_1, \frac{\partial\omega}{\partial t}\right)\frac{\partial\omega}{\partial t}\right\rangle \mathrm{d}t.$$

Then RHS is independent of α so E_{**} is well-defined and bilinear for geodesics.
Proof: Recall

$$\frac{1}{2}\frac{\partial E}{\partial u_2} = -\sum_t \left\langle \frac{\partial\alpha}{\partial u_2}, \Delta_t \frac{\partial\alpha}{\partial t}\right\rangle - \int_0^1 \left\langle \frac{\partial\alpha}{\partial u_2}, \frac{\partial}{\partial t}\frac{\partial\alpha}{\partial t}\right\rangle \mathrm{d}t.$$

Taking derivatives, we obtain

$$\frac{1}{2}\frac{\partial^2 E}{\partial u_1 \partial u_2} = -\sum_t \left\langle \frac{\partial^2 \alpha}{\partial u_1 \partial u_2}, \Delta_t \frac{\partial \alpha}{\partial t} \right\rangle - \sum_t \left\langle \frac{\partial \alpha}{\partial u_2}, \frac{\partial}{\partial u_1} \Delta_t \frac{\partial \alpha}{\partial t} \right\rangle \tag{11.1}$$

$$- \int_0^1 \left\langle \frac{\partial^2 \alpha}{\partial u_1 \partial u_2}, \frac{\partial^2 \alpha}{\partial t^2} \right\rangle \mathrm{d}t - \int_0^1 \left\langle \frac{\partial \alpha}{\partial u_2}, \frac{\partial}{\partial u_1} \frac{\partial^2 \alpha}{\partial t^2} \right\rangle \mathrm{d}t. \tag{11.2}$$

Since ω is a geodesic, the first and third terms are zero.
Hence

$$\frac{1}{2}\frac{\partial^2 E}{\partial u_1 \partial u_2} = -\sum_t \left\langle \frac{\partial \alpha}{\partial u_2}, \frac{\partial}{\partial u_1} \Delta_t \frac{\partial \alpha}{\partial t} \right\rangle - \int_0^1 \left\langle \frac{\partial \alpha}{\partial u_2}, \frac{\partial}{\partial u_1} \frac{\partial^2 \alpha}{\partial t^2} \right\rangle \mathrm{d}t.$$

Where

$$\frac{\partial}{\partial u_1} \cdot \frac{\partial}{\partial t} \left(\frac{\partial \alpha}{\partial t} \right) - \frac{\partial}{\partial t} \frac{\partial}{\partial u_1} \left(\frac{\partial \alpha}{\partial t} \right) = R\left(W_1, \frac{\partial \alpha}{\partial t} \right) \frac{\partial \alpha}{\partial t}. \left(\text{Since } \left[\frac{\partial}{\partial t}, \frac{\partial}{\partial u_1} \right] = 0 \right)$$

and $\frac{\partial}{\partial t}\left(\frac{\partial}{\partial u_1}\frac{\partial \alpha}{\partial t}\right) = \frac{\partial}{\partial t}\left(\frac{\partial}{\partial t}\frac{\partial \alpha}{\partial u_1}\right) = \frac{\partial^2}{\partial t^2} W_1$.

Caution: The first variation formula holds for all paths, not only geodesics, but the second variation formula holds only for geodesics (critical points).

12

Jacobi Fields

2026/5/21

Milnor *Morse Theory* Chapter14: Jacobi Fields: The Null Space of E_{**}.

Jacobi fields

Recall the second variation formula

$$\frac{1}{2}\frac{\partial^2 E}{\partial u_1 \partial u_2}(0,0) = -\sum_t \left\langle W_2(t), \Delta_t \frac{\partial W_1}{\partial t} \right\rangle - \int_0^1 \left\langle W_2, \frac{\partial^2 W_1}{\partial t^2} + \mathcal{R}\left(W_1, \frac{\partial \omega}{\partial t}\right)\frac{\partial \omega}{\partial t} \right\rangle \mathrm{d}t.$$

We know that a geodesic is a critical point, but this alone does not imply it is minimal.
This motivates us to define: A vector field J along ω is called a **Jacobi field** if it satisfies the Jacobi differential equation

$$\frac{\partial^2 J}{\partial t^2} + \mathcal{R}\left(J, \frac{\partial \omega}{\partial t}\right)\frac{\partial \omega}{\partial t} = 0. \qquad (\star)$$

This is a second order linear ODE, so the solution is uniquely determined by the initial conditions $J(0), \frac{\partial J}{\partial t}(0) \in T_{\omega(0)}M$, and all solutions are smooth.

Let $p = \omega(0), q = \omega(1)$ be two points on the geodesic ω, $T\Omega_\omega$ denotes the vector fields along ω vanishing at p, q.

Recall the null space of $E_{**} : T\Omega_\omega \times T\Omega_\omega \to \mathbb{R}$ is the kernel of $\tilde{E} : T\Omega_\omega \to T\Omega_\omega^*, W_1 \mapsto E_{**}(W_1, \cdot)$.
The nullity ν of E_{**} is the dimension of the null space and E_{**} is degenerate if $\nu > 0$.

Theorem: A vector field $W_1 \in T\Omega_\omega$ belongs to the null space iff W_1 is a Jacobi field. (But a Jacobi field is likely to be not in $T\Omega_\omega$.)
Proof: If J is a Jacobi field vanishing at p, q, then $J \in T\Omega_\omega$ and the second variation formula implies

$$-\frac{1}{2}E_{**}(J, W_2) = \sum \langle W_2(t), 0\rangle + \int_0^1 \langle W_2, 0\rangle \,\mathrm{d}t = 0.$$

Hence J belongs to the null space.
Conversely, suppose W_1 is in the null space. Choose a subdivision $0 = t_0 < \cdots < t_k = 1$ such that $W_1|_{[t_{i-1}, t_i]}$ is smooth, and let $f : [0,1] \to [0,1]$ be smooth such that $f(t_i) = 0$ and otherwise $f(x) > 0$.

Let

$$W_2(t) = f(t)\left(\frac{\partial^2 W_1}{\partial t^2} + \mathcal{R}(W_1, V)V\right).$$

Then

$$0 = -\frac{1}{2}E_{**}(W_1, W_2) = \int_0^1 f(t)\|\frac{\partial^2 W_1}{\partial t^2} + \mathcal{R}(W_1, V)V\|^2\,\mathrm{d}t.$$

Therefore W_1 satisfy $(\star)$ on $x \neq t_i$.
Now let $W_2' \in T\Omega_\omega$ be a field such that $W_2'(t_i) = \Delta_{t_i}\frac{\partial W_1}{\partial t}$ for $i = 1, 2, \cdots, k-1$, then

$$0 = -\frac{1}{2}E_{**}(W_1, W_2') = \sum_{i=1}^{k-1}\|\Delta_{t_i}\frac{\partial W_1}{\partial t}\|^2$$

so $\frac{\partial W_1}{\partial t}$ is continuous. By ODE theory, a C^1 solution of $(\star)$ is smooth. $\square$

It follows that the nullity ν is finite, since there are at most n linearly independent Jacobi fields along ω given the boundary condition $J(0) = 0$. Actually $\nu < n$, since we can find $J(0) = 0$ but $J(1) \neq 0$: Let $J(t) = tV(t)$ where $V = \frac{\partial \omega}{\partial t}$ is the velocity, then $\frac{\partial J}{\partial t} = V + t\frac{\partial V}{\partial t} = V$ since $\frac{\partial V}{\partial t} = 0$. Hence $\frac{\partial^2 J}{\partial t^2} = 0$. Also, $\mathcal{R}(J, V)V = t\mathcal{R}(V, V)V = 0$. Therefore J is a Jacobi field with $J(0) = 0$ but $J(1) = V(1) \neq 0$.

Example: Suppose M is "flat" such that $\mathcal{R} \equiv 0$, then the Jacobi equation becomes $\frac{\partial^2 J}{\partial t^2} = 0$. Suppose $J = \sum f^i P_i$ where P_i are parallel along ω, then the equation is $\frac{\partial^2}{\partial t^2} f_i = 0$ so each f_i is linear. Hence any Jacobi field has at most one zero, so E_{**} is non-degenerate.

Let α be a variation of ω, (not necessarily keeping endpoints fixed), such that $\alpha(u, \cdot)$ is a geodesic for all $u \in (-\varepsilon, \varepsilon)$ and $\alpha(0, \cdot) = \omega$.
Lemma: If α is such a variation of ω through geodesics, then the variation vector field $W(t) = \frac{\partial \alpha}{\partial u}(0, t)$ is a Jacobi field along ω.
Proof: Note that

$$0 = \frac{\partial}{\partial u}\left(\frac{\partial}{\partial t}\frac{\partial \alpha}{\partial t}\right) = \frac{\partial}{\partial t}\frac{\partial}{\partial u}\frac{\partial \alpha}{\partial t} + \mathcal{R}\left(\frac{\partial \alpha}{\partial u}, \frac{\partial \alpha}{\partial t}\right)\frac{\partial \alpha}{\partial t} \tag{12.1}$$

$$= \frac{\partial^2}{\partial t^2}\frac{\partial \alpha}{\partial u} + \mathcal{R}\left(\frac{\partial \alpha}{\partial u}, \frac{\partial \alpha}{\partial t}\right)\frac{\partial \alpha}{\partial t}. \tag{12.2}$$

Hence evaluate at $(u, t) = (0, t)$ we infer W is a Jacobi field. $\square$
Thus one way of obtaining Jacobi fields is moving geodesics around (an exponential map $\alpha(u, t) = \exp_p tv(u)$). The converse is also true:
Lemma: Every Jacobi field along a geodesic $\omega : I \to M$, with $J(0) = 0$, may be obtained by a variation of ω through geodesics.
Proof: If J is a Jacobi field along ω with $J(0) = 0$, then let $w = \frac{\partial J}{\partial t}(0)$ and $v = \frac{\partial \omega}{\partial t}(0)$. Take $v(s) : I \to T_{\omega(0)}M$ such that $v(0) = v, v'(0) = w$ and let $\alpha(u, t) = \exp_p(tv(u))$, then $W = \frac{\partial \alpha}{\partial u}(0, t)$ is a Jacobi field such that $W(0) = J(0) = 0$ and $\frac{\partial W}{\partial t}(0) = \frac{\partial J}{\partial t}(0) = w$. Hence by uniqueness $W = J$.

Conjugate points

Definition: p, q are **conjugate** along ω if there exists a non-zero Jacobi field J along ω which vanishes for $t = a, b$. The **multiplicity** of p, q is equal to the dimension of all such Jacobi fields as an $\mathbb{R}-$vector

space.
Then the Theorem states that: E_{**} is degenerate iff the end points p, q are conjugate along ω; and ν is equal to the multiplicity of p, q.

Example: Suppose p, q are antipodal points on $\mathbb{S}^n$, and ω a great circle arc from p to q. Using Lemma, we can rotate the sphere in $n-1$ directions, fixing p, q, to obtain $n-1$ linearly independent Jacobi fields. Hence p, q are conjugate of multiplicity $n-1$, so the nullity ν takes its maximal value.
So conjugation shows how the geodesic fails to be of minimal energy.

More of this topic is discussed in the next chapter.

13

Fundamental Theorem of Morse Theory

2026/5/21

Morse Index Theorem

Let $\Omega(M,p,q)$ be piece-wise smooth paths from p to q, and $\Omega^*(M,p,q)$ be smooth paths.
A **path** in $\Omega(M,p,q)$ is a variation of path ω, which corresponds to a vector field on ω.

Main Theorem: (**Morse Index Theorem**) The index λ of E_{**} is equal to the number of points $\gamma(t), 0 < t < 1$ such that $\gamma(t)$ is conjugate to $\gamma(0)$ along γ, each point counted with its multiplicity. Furthermore, λ is always finite.
Corollary: There is only a finite number of conjugate points along a compact geodesic segment.

Choose a subdivision $0 = t_0 < \cdots < t_k = 1$ such that $\gamma|_{[t_{i-1},t_i]}$ is minimal.
Let $T\Omega_\gamma(t_0,\cdots,t_k)$ be a subspace of $T\Omega_\gamma$, consisting of W along γ such that: (a) $W|_{[t_{i-1},t_i]}$ is a Jacobi field along $\gamma|_{[t_{i-1},t_i]}$ and (b) W vanishes on $0, 1$.
Let T' be the subspace consisting of W along γ such that $W(t_i) = 0$.

Lemma: $T\Omega_\gamma$ splits as orthogonal (with the bilinear form E_{**}) direct sum $T\Omega_\gamma(t_0,t_1,\cdots,t_k) \oplus T'$, and $E_{**}|_{T'}$ is positive definite.
Proof: Given $W \in T\Omega_\gamma$, there is a unique Jacobi field $W_1 \in T\Omega_\gamma(t_0,\cdots,t_k)$ such that $W_1(t_i) = W(t_i)$ (by ODE theory). Clearly $W - W_1 \in T'$ so $T\Omega_\gamma = T\Omega_\gamma(t_0,\cdots,t_k) \oplus T'$ as direct sums, and by the second variation formula they are orthogonal.
For $W \in T\Omega_\gamma$, by definition, $E_{**}(W,W) = \frac{\partial^2}{\partial u^2}(E\circ\alpha)(0)$ where $\alpha : (-\varepsilon,\varepsilon)\times I \to \Omega$ is a variation with $\frac{\partial\alpha}{\partial u}(0) = W$. If $W \in T'$ we may assume α is a variation fixing $\alpha(u,t_i)$.
It suffices to show E_{**} is positive definite on T':
For each $W \in T'$, any variation α is piece-wise smooth, while γ is minimal on $[t_{i-1},t_i]$. Hence $E(\alpha(u,\cdot)) \geqslant E(\alpha(0,\cdot)) = E(\gamma)$ and we have a local minimum, so the second derivative $E_{**}|_{T'}(W,W) \geqslant 0$.
If $E_{**}(W,W) = 0$ for $W \in T'$ we show that W is in the null space: we already have $E_{**}(W,W') = 0 \forall W' \in T\Omega_\gamma(t_0,\cdots,t_k)$; for $W' \in T'$, note that

$$0 \leqslant E_{**}(W+cW', W+cW') = 2cE_{**}(W,W') + c^2E_{**}(W',W')$$

for all c, which implies $E_{**}(W,W') = 0$. Hence W should be a Jacobi field, so $W = 0$. □

Therefore, the index and nullity of E_{**} on $T\Omega_\gamma$ is reduced to the index and nullity of $E_{**}|_{T\Omega_\gamma(t_0,\cdots,t_k)}$, where $T\Omega_\gamma(t_0,\cdots,t_k)$ is finite dimensional, since the space of Jacobi fields is of finite dimension.

For any $\tau \in [0,1]$, let $(E_0^\tau)_{**}$ be the Hessian of $\gamma_\tau = \gamma|_{[0,\tau]}$, and $\lambda(\tau)$ be the index of $(E_0^\tau)_{**}$. It suffices to prove the following:

- λ is monotone increasing: easy since vector fields can be extended.
- λ is left continuous: suppose $t_i < \tau < t_{i+1}$, then $\lambda(r)$ is the index of E_{**} restricted on the space of broken Jacobi field under the subdivision $0 < t_1 < \cdots < t_i < \tau$. Since any broken Jacobi field is determined by its values at endpoints, it is isomorphic to the direct sum $\Sigma = TM_{\gamma(t_1)} \oplus \cdots \oplus TM_{\gamma(t_i)}$ which is independent of τ, and the quadratic form H_τ on Σ varies continuously with τ. Therefore if H_τ is negative definite on a subspace V, then for nearby τ', $H_{\tau'}$ is also negative definite on V, so $\lambda(\tau') \geqslant \lambda(\tau)$. Hence λ is left-continuous (since it is monotone).
- $\lambda(\tau) = 0$ for small τ: since γ is a minimal geodesic, there are no conjugate points.
- If the nullity of $(E_0^\tau)_{**}$ is ν, then $\lambda(\tau + \varepsilon) = \lambda(\tau) + \nu$ for small ε.

A finite dimensional approximation to Ω^c

Consider $\Omega = \Omega(M; p, q)$. Let ρ define the topological metric coming from the Riemannian metric, and

$$d(\omega, \omega') = \max \rho(\omega(t), \omega'(t)) + \|\frac{ds}{dt} - \frac{ds'}{dt}\|_2.$$

(The L^2 integral term is to let $E_a^b(\omega)$ be continuous for b.)
Let $\Omega^c = E^{-1}([0,c])$, $\mathrm{int}\Omega^c = E^{-1}([0,c))$.
Choose a subdivision $0 = t_0 < \cdots < t_k = 1$. Let $\Omega(t_0, \cdots, t_k)$ be the subspace consisting of $\omega : [0,1] \to M$ such that $\omega(0) = p, \omega(1) = q$ such that $\omega|_{[t_{i-1}, t_i]}$ is a geodesic. $\Omega^c(t_0, \cdots, t_k) = \Omega^c \cap \Omega(t_0, \cdots, t_k)$ and likewise $\mathrm{Int}\Omega^c(t_0, \cdots, t_k)$.

Lemma: M complete Riemannian manifold, given $c > 0$ and $\Omega^c \neq \emptyset$, for sufficiently fine divisions $t_0, \cdots, t_k$, $\mathrm{Int}\Omega^c(t_0, \cdots, t_k)$ can be given the structure of a finite dimensional smooth manifold.
Proof: Let $S = \{x \in M : \rho(x,p) \leqslant \sqrt{c}\}$, then S is compact (since M is complete, use Hopf-Rinow) and every path $\omega \in \Omega^c$ lies in S.
Hence there exists $\varepsilon > 0$ such that $x, y \in S$ and $\rho(x,y) < \varepsilon$ implies there is a unique geodesic connecting x, y with length $< \varepsilon$.
For any subdivision such that $t_i - t_{i-1} < \varepsilon^2/c$, we know that any broken geodesic $\omega \in \Omega^c(t_0, \cdots, t_k)$ satisfy

$$(L_{t_{i-1}}^{t_i} \omega)^2 = (t_i - t_{i-1}) E_{t_{i-1}}^{t_i} \omega \leqslant \varepsilon^2.$$

Hence $\omega|_{[t_{i-1}, t_i]}$ is unique, so ω is uniquely determined by its values at $t_1, \cdots, t_{k-1}$. Therefore we have a homeomorphism $\omega \mapsto (\omega(t_1), \cdots, \omega(t_{k-1}))$ from $\mathrm{Int}\Omega^c(t_0, \cdots, t_k)$ to an open subset of M^{k-1}.

Denote $B = \mathrm{Int}\Omega^c(t_0, \cdots, t_k)$, and let $E' : B \to \mathbb{R}$ be the restriction of the energy function.
Theorem: $E' : B \to \mathbb{R}$ is smooth and for any $a < c$, $B^a = (E')^{-1}([0,a])$ is compact and a deformation retraction of Ω^a. The critical points of E' are the critical points of E in $\mathrm{Int}\Omega^c$, and their index/nullity are the same.
Proof: The "broken geodesic" $\omega \in B$ depends smoothly on the tuple $\omega(t_1), \cdots, \omega(t_{k-1})$, hence so does the energy $E'(\omega)$:

$$E'(\omega) = \sum_{j=1}^{k} \frac{\rho(\omega(t_{j-1}), \omega(t_j))^2}{t_j - t_{j-1}}.$$

As a closed subset of a compact set, B^a is also compact for $a < c$.

Consider the deformation retraction $r : \mathrm{Int}\Omega^c \to B$ as follows: for $\omega \in \mathrm{Int}\Omega^c$, let $r(\omega) \in B$ be the unique broken geodesic such that each $r(\omega)|_{[t_{i-1},t_i]}$ is a geodesic of length $< \varepsilon$ connecting $\omega(t_{i-1}), \omega(t_i)$.
Clearly $E(r(\omega)) \leqslant E(\omega) < c$ and this retraction has a homotopy r_u defined by

$$\begin{cases} r_u(\omega)|_{[0,t_{i-1}]} & = r(\omega)|_{[0,t_{i-1}]}, \\ r_u(\omega)|_{[t_{i-1},u]} & = \text{minimal geodesic connecting } \omega(t_{i-1}), \omega(u), \\ r_u(\omega)|_{[u,1]} & = \omega|_{[u,1]}. \end{cases}$$

where $t_{i-1} \leqslant u < t_i$. Since this homotopy does not increase energy, B^a is also a deformation retraction of Ω^a.
Every geodesic is a broken geodesic, so the critical points of E are also critical points of E'. By the first variation formula, the critical points of E' are exactly those of E.
The tangent space TB_γ of a broken geodesic $\gamma \in B$ can be identified with the space $T\Omega_\gamma(t_0, \cdots, t_k)$ of broken Jacobi fields along γ: for any variation $\alpha : (-\varepsilon, \varepsilon) \to B$ which represents a tangent vector, the vector field $\frac{\partial \alpha}{\partial u}(0,t)$ is a Jacobi field along γ. By the corollary in the previous chapter, the index of E and E' are the same.

Corollary: Let M be a complete Riemannian manifold, and let $p, q \in M$ be two points that are not conjugate along any geodesic of length $\leqslant \sqrt{a}$. Then Ω^a has homotopy type of a finite CW complex, with a $\lambda-$cell for each critical point of E in Ω^a with index λ.

Topology of Full Path Space

Now consider $\Omega^*(M; p, q)$ be all continuous paths from p to q, with the compact open topology. This topology is also induced by the metric

$$d^*(\omega, \omega') = \max \rho(\omega(t), \omega'(t)).$$

Since $d \geqslant d^*$, the map $\iota : \Omega \hookrightarrow \Omega^*$ is continuous.

Theorem: ι is a homotopy equivalence.

Fundamental theorem of Morse Theory: Let M be a complete Riemannian manifold, $p, q \in M$ not conjugate along any geodesic, then $\Omega(M; p, q)$ and $\Omega^*(M; p, q)$ have homotopy type of a countable CW-complex, with one cell of dim λ for each geodesic $p \to q$ of index λ.
Proof: Take a subdivision $a_0 < a_1 < \cdots$ where a_i are not critical points of E, and each (a_i, a_{i+1}) has exactly one critical point.

Example: Path space of $\mathbb{S}^n$: Suppose p, q are non-conjugate points of $\mathbb{S}^n$, i.e. $q \neq p, p'$ where p' is the antipodal point of p. Then there are countably many geodesics of p, q, e.g. γ_0 be the shortest great circle arc pq, γ_1 be the long great circle $pq'p'q$, etc. The index of γ_k is $k(n-1)$ since each p, p' in the geodesic is conjugate to p with multiplicity $n-1$.
Therefore, the loop space $\Omega(\mathbb{S}^n)$ has the homotopy type of a CW complex with one cell in each dimensions $0, n-1, 2(n-1), \cdots$.
As a corollary, let M has homotopy type $\mathbb{S}^n$, then for $n > 2$ by calculating homology, any two non-conjugate points of M are joined by infinitely many geodesics.

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Topology and Curvature

2026/5/29

(Notation differs from the textbook)
Recall $\mathcal{R}(U,V)W = (D_U D_V - D_V D_U - D_{[U,V]})W$ and $\mathcal{R}(U,V,W,Z) = \langle \mathcal{R}(U,V)W, Z\rangle$. (consider the 3-dim case R_{1212}).

Part0: Ricci/Sectional curvature

Sectional curvature: $K(u,v) = -\mathcal{R}(u,v,u,v)/\|u\wedge v\|^2$. (For $u,v \in T_pM$, consider $\exp_p : T_pM \to M$ which sends $\langle u,v\rangle$ to a 2-dim subspace of M. The Gaussian curvature of this subspace is exactly $K(u,v)$.)
Note that $\mathcal{R}(u,v,w,z) = \mathcal{R}(w,z,u,v) = -\mathcal{R}(v,u,w,z)$, so K completely determines $\mathcal{R}$ (by linear algebra).
Ricci curvature: (for orthonormal e_i)

$$\operatorname{Ric}(u,v) = \operatorname{tr}(\mathcal{R}(-,u)v) = \sum_{i,j} g^{ij}\langle \mathcal{R}(e_i,u)v, e_j\rangle = \sum_{i,j} \mathcal{R}(e_i,u,v,e_j).$$

Hence $\operatorname{Ric}(e_i,e_i) = \mathcal{R}(e_i,e_j,e_j,e_i) = -\sum_j K(e_i,e_j)$.
If we write in tensors, $K(e_i,e_j) = \mathcal{R}_{ijij}$, and $\operatorname{Ric}(e_i,e_i) = -\sum_j \mathcal{R}_{ijij}$.

Part1: Revisit Exp map & Jacobi fields

Existence of non-conjugate points: We show that for any point a there is a small neighborhood U such that any b is not conjugate to a along the unique short geodesic $\omega : a \to b$ in U.
Consider $\Phi(t,s) = \exp_p t(v+sw)$, $V = \frac{\partial \Phi}{\partial t} = \exp_{p*}|_{tv} v$ and $J = \frac{\partial\Phi}{\partial s} = t\exp_{p*}|_{tv} w$. Then $\ddot{J} = D_V D_V J = D_V D_J V = \mathcal{R}(V,J)V$ so J is a Jacobi field such that $J(0)=0$, $\dot{J}(0) = w$ and $J(1) = \exp_{p*}|_v w$. Since $\dot{J}(0) = w$ gives all Jacobi fields along ω, it suffices to show $\exp_{p*}|_v$ is non-singular, which is clear by $\exp_p$ is a local diffeomorphism.

Theorem (Cartan isometry): Given an isometry $L : T_pM \to T_qN$, if for all u,v tangent vectors in an exponential map $\exp_p : B_r(p) \to M$, we have $K(u,v) = K(Lu,Lv)$. Then L extends to a local isometry f such that $f(p) = q$ and $df_p = L$, given by $f = \exp_q \circ L \circ \exp_p^{-1}$.

Part2: Negative curvature, Cartan-Hadamard

Theorem: M simply-connected complete Riemannian manifold, with non-positive sectional curvature. Then $\exp_p : T_pM \to M$ is a diffeomorphism, and any two points of M is connected by a unique geodesic.
Proof: Use the lemma below. By index theorem, $\Omega(p,q)$ is equivalent to zero cells as a CW-complex. Since $\Omega(p,q)$ is connected (by M simply connected), $\Omega(p,q) \simeq$ pt. Completeness gives $\exp_p$ surjective, and $\Omega(p,q) \simeq$ pt gives injective, since p, q are non-conjugate implies geodesics are discrete, so connectedness implies there is only one 0-cell (geodesic).
Lemma: If $K \leqslant 0$ then there are no conjugate points.
Proof: Since $\ddot{J} = R(V,J)V$, $\frac{d}{dt}\langle \dot{J}, J\rangle = \langle \ddot{J}, J\rangle + |\dot{J}|^2 = \mathcal{R}(V,J,V,J) + |\dot{J}|^2 \geqslant 0$. If $J(0) = J(1) = 0$ then $\langle \dot{J}, J\rangle \equiv 0$ so $\dot{J} \equiv 0$ and there are no conjugate points.

Part3: Positive curvature, Bonnet-Myers

Theorem: If M complete n-dimensional, and $\mathrm{Ric}(u,u) \geqslant \frac{n-1}{r^2} \forall u$ everywhere, then M is compact and $\mathrm{diam}(M) \leqslant \pi r$.
Proof: We compare M with S^n. Let $\gamma : [0,1] \to M$ be a geodesic with $L(\gamma) = L$. Take orthonormal parallel vector fields $P_1, \cdots, P_n$ along γ such that P_n points along γ, then $V = \frac{d\gamma}{dt} = L \cdot P_n$, $\frac{DP_i}{dt} = 0$.
Let $W_i(t) = \sin(\pi t) P_i(t)$, then

$$\frac{1}{2}E_{**}(W_i, W_i) = -\int_0^1 \left\langle W_i, \frac{\partial^2 W_i}{\partial t^2} + \mathcal{R}(V, W_i)V \right\rangle \mathrm{d}t = \int_0^1 \sin^2(\pi t)(\pi^2 - L^2 \langle \mathcal{R}(P_n, P_i)P_n, P_i\rangle)\,\mathrm{d}t.$$

Summing up $i = 1, \cdots, n-1$, we obtain

$$\frac{1}{2}\sum E_{**}(W_i, W_i) = \int_0^1 \sin^2(\pi t)((n-1)\pi^2 - L^2 \mathrm{Ric}(P_n, P_n))\,\mathrm{d}t.$$

If $\mathrm{Ric}(P_n, P_n) \geqslant \frac{n-1}{r^2}$ and $L > \pi r$, then this expression is negative, hence $E_{**}(W_i, W_i) < 0$ for some i, so the index of γ is positive, and by the index theorem, γ contains conjugate points. Hence γ is not a minimal geodesic, and since M is complete, any two points has distance $\leqslant \pi r$.

15

Compact Lie Groups

2026/6/4

Milnor *Morse Theory* Chapters 20,21,22

Symmetric Space

Def: A symmetric space is a connected Riemannian manifold such that for each $p \in M$ there is an isometry $I_p : M \to M$ fixing p and reversing geodesics through p.

Lemma: Let γ be a geodesic on $M, p = \gamma(0), q = \gamma(c)$. Then $I_q I_p(\gamma(t)) = \gamma(t+2c)$ if $\gamma(t), \gamma(t+2c)$ are defined. Moreover, $I_q I_p$ preserves parallel vector fields along γ.
Proof: $I_q I_p(\gamma(t)) = \gamma(t+2c)$ is trivial. Notice that for any V parallel along γ, $(I_p)^*(V)$ is also parallel, and $(I_p)^*(V(0)) = -V(0)$. Hence $(I_p)^*(V(t)) = -V(-t)$, and $(I_q)^*(I_p)^*(V(t)) = V(t+2c)$ is still parallel.

Corollary: (1) M is complete, since by Lemma any geodesic can be infinitely extended. (2) I_p is unique, since any point $q \in M$ is joined to p by a geodesic. (3) If U, V, W are parallel along γ then $\mathcal{R}(U, V)W$ is parallel.
Proof of (3): For any $p = \gamma(0), q = \gamma(c)$, put $T : I_{\gamma(c/2)} \circ I_p$ and X_q and parallel vector field along γ, then

$$\langle \mathcal{R}(U_q, V_q)W_q, X_q\rangle = \langle \mathcal{R}(T^*U_p, T^*V_p)T^*W_p, T^*X_p\rangle = \langle \mathcal{R}(U_p, V_p)W_p, X_p\rangle$$

so it is constant for any X. Hence $\mathcal{R}(U, V)W$ is parallel.
(If only (3) holds, M is called locally symmetric.)

Def: Let $\gamma : \mathbb{R} \to M$ be a geodesic in a (locally) symmetric space, and $V = \frac{d\gamma}{dt}(0)$. Define $K_V : TM_p \to TM_p, W \mapsto \mathcal{R}(V, W)V$. The K_V is self-adjoint:

$$\langle K_V(W), W'\rangle = \langle \mathcal{R}(V, W)V, W'\rangle = \langle \mathcal{R}(V, W')V, W\rangle = \langle W, K_V(W')\rangle.$$

Hence we can choose orthonormal U_i such that $K_V U_i = e_i U_i$.
Theorem: The conjugate points to $p = \gamma(0)$ along γ are the points $\gamma(\pi k/\sqrt{e_i})$ where $k \in \mathbb{Z}\backslash\{0\}$ and e_i is a negative eigenvalue of K_V. The multiplicity of $\gamma(t)$ is equal to the number of e_i such that $t = k\pi/\sqrt{-e_i}$.
Proof: Extend U_i to vector fields along γ, then since M is locally symmetric, $\mathcal{R}(V, U_i)V = e_i U_i$ everywhere along γ.

Any vector field W can be written as $W(t) = \sum w_i(t)U_i(t)$. Then the Jacobi equation becomes

$$0 = \frac{\partial^2}{\partial t^2}W + K_V(W) = \sum \frac{\partial^2 w_i}{\partial t^2}U_i + \sum e_i w_i U_i \iff \frac{\partial^2}{\partial t^2}w_i(t) = -e_i w_i(t).$$

Hence if $e_i < 0$ then the solutions are $c_i \sin(\pi t/\sqrt{-e_i})$, and if $e_i > 0$ they are $c_i \sinh(\pi t/\sqrt{e_i})$ which does not vanish.

Lie Groups

Denote $L_a(g) = ag$ and $R_a(g) = ga$.
First we construct a left and right invariant Haar measure on a compact Lie group. For a point p, take an n-form $\omega \in \Lambda^n(T_pM)$, push ω to be a left-invariant n-form on M.
Fact: Left invariant n-forms are also right invariant.
Proof: Note that $(R_a)^*\omega$ is also left-invariant, since $(L_b)^*(R_a)^*\omega = (R_a)^*(L_b)^*\omega = (R_a)^*\omega$. We can define $(R_a)^*\omega = f(a)\omega$, then f defines a Lie group homomorphism $M \to \mathbb{R}^\times$. Since M is connected and compact, it is bounded, and since $f(g^n) = f(g)^n$, we have $f(g) \equiv 1$.
Define a left Haar measure to be $\mu(E) = \int_E |\omega|$, then since ω is left and right invariant, so is the Haar measure.
Then we can define a left&right invariant Riemannian metric, from an arbitrary metric:

$$\langle\langle V, W\rangle\rangle = \int_{G\times G} \langle (L_g)^*(R_h)^*(V), (L_g)^*(R_h)^*(W)\rangle \, \mathrm{d}\mu(g)\mathrm{d}\mu(h).$$

Lemma: If G is a Lie group with bi-invariant metric, then G is a symmetric space, and the reflection is $I_g(\sigma) = g\sigma^{-1}g$.
Proof: We consider $I_e : g \mapsto g^{-1}$, write $m : G \times G \to G, (g,h) \mapsto gh$ then $dm|_{(e,e)}(X,Y) = X + Y$. The composition $f : G \xrightarrow{(id\times inv)} G \times G \xrightarrow{m} G$ is constant, so $0 = df|_e$ implies $d\mathrm{inv}|_e = -id$.
Furthermore, $(R_{\sigma^{-1}})_* \circ (inv)_* \circ (L_{\sigma^{-1}})_*$ is an isometry from TG_σ to $TG_{\sigma^{-1}}$. Hence we can write $I_g = L_g \circ I_e \circ L_{g^{-1}}$.

Lemma: The geodesics γ in G with $\gamma(0) = e$ are the Lie group homomorphisms (1-parameter subgroup) $\mathbb{R} \to G$.
Proof: $I_{\gamma(t)}I_e$ sends $\gamma(u)$ to $\gamma(u+2t)$, and $I_{\gamma(t)}I_e(\sigma) = \gamma(t)\sigma\gamma(t)$, so $\gamma(t)\gamma(u)\gamma(t) = \gamma(u+2t)$. Hence $(\gamma(t))^n = \gamma(nt)$ so γ is a homomorphism.

Def: A Lie algebra is an algebra satisfying

$$[[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0.$$

Example: The left invariant vector fields of a Lie group is its Lie algebra.

Theorem: Given X, Y, Z, W left-invariant vector fields on G, then
(1) $\langle [X,Y], Z\rangle = \langle X, [Y,Z]\rangle$; (2) $\mathcal{R}(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$;
(3) $\langle \mathcal{R}(X,Y)Z, W\rangle = -\frac{1}{4}\langle [X,Y],[Z,W]\rangle$.
Proof: Note that the tangent vector field along a geodesic is left invariant, so we have $D_X X = 0$ for any left-invariant vector field X.
$0 = D_{(X+Y)}(X+Y) = D_X X + D_Y Y + D_X Y + D_Y X$, so $2D_X Y = [X,Y]$.
$0 = D_Y\langle X, Z\rangle = \langle D_Y X, Z\rangle + \langle X, D_Y Z\rangle = \frac{1}{2}\langle [Y,X], Z\rangle + \frac{1}{2}\langle X, [Y,Z]\rangle$, so we have (1).
(2)

$$\mathcal{R}(X,Y)Z = D_X D_Y Z - D_Y D_X Z - D_{[X,Y]}Z = \frac{1}{4}[X,[Y,Z]] - \frac{1}{4}[Y,[X,Z]] - \frac{1}{2}[[X,Y],Z] \tag{15.1}$$

$$= -\frac{1}{4}[[X,Y],Z] \tag{15.2}$$

(3) by (2) and (1),

$$\langle \mathcal{R}(X,Y)Z, W\rangle = -\frac{1}{4}\langle [[X,Y],Z],W\rangle = -\frac{1}{4}\langle [X,Y],[Z,W]\rangle.$$

Corollary: The sectional curvature $K(X,Y) = -\langle \mathcal{R}(X,Y)X,Y\rangle/\|X\wedge Y\|^2$ is positive, since $\langle \mathcal{R}(X,Y)X,Y\rangle = -\frac{1}{4}\langle [X,Y],[X,Y]\rangle$.
Corollary: If G has bi-invariant metric, and the Lie algebra $\mathfrak{g}$ has trivial center (=$\{g\in\mathfrak{g} : [g,\cdot]\equiv 0\}$), then G is compact with finite fundamental group.
Corollary (hard): A simply connected Lie group G with bi-invariant metric splits as a Cartesian product $G'\times\mathbb{R}^n$ where G' is compact with trivial center.

Main Theorem: (Bott) Let G be a simply connected Lie group with bi-invariant metric, then the loop space has a homotopy type of a CW-complex with no odd-dimensional cells, and if G is compact, only finitely many $\lambda-$cells for each λ.
Proof: (1) We show that the index of any geodesic is always even, then by Morse Index Theorem there are no odd-dim cells.
Recall the conjugate points of p on γ are determined by the eigenvalues of $K_V : W \mapsto \mathcal{R}(V,W)V$ where $V = \frac{d\gamma}{dt}(0)$. We showed that $\mathcal{R}(V,W)V = -\frac{1}{4}[[V,W],V]$. Define $\mathrm{ad}_V : \mathfrak{g}\to\mathfrak{g}, W\mapsto [V,W]$ then $K_V = \frac{1}{4}\mathrm{ad}_V\circ\mathrm{ad}_V$.
Note that ad_V is skew-symmetric: $\langle [V,W],W'\rangle = \langle -[W,V],W'\rangle = -\langle W,[V,W']\rangle$. Hence under a proper orthogonal basis ad_V has the form $\mathrm{diag}(\begin{pmatrix} 0 & a_1 \\ -a_1 & 0\end{pmatrix},\cdots)$ and K_V has the form $\mathrm{diag}(-a_1^2,-a_1^2,-a_2^2,\cdots)$.
Therefore the index of any geodesic is even.
(2) Since G is compact and simply-connected, $\mathfrak{g}$ has trivial center, using the corollary above. Take an orthonormal basis $X_1,\cdots,X_n$ then $\mathcal{R}(X_i,X_i) = \sum K(X_i,X_j) = \sum\frac{1}{4}\|[X_i,X_j]\|^2 > 0$. Since G is compact, $\mathcal{R}(u,u)$ has a positive lower bound, so by Myers theorem, if a geodesic has length $\geqslant k\pi r$ then the index of it is $\geqslant k$.

Whole Manifolds of Minimal Geodesics

Main Theorem: If the space Ω^d of minimal geodesics from p to q is a topological manifold, and if every non-minimal geodesic from p to q has index $\geqslant\lambda$, then the relative homotopy $\pi_i(\Omega,\Omega^d)$ is zero for $i<\lambda$.
Corollary: (Freudenthal suspension theorem) $\pi_i(\mathbb{S}^n)\cong\pi_{i+1}(\mathbb{S}^{n+1})$ for $i\leqslant 2n-2$.

Proof of Main Theorem: Consider the energy function $E:\Omega\to\mathbb{R}$, and subspace $\Omega^c = E^{-1}([0,c))$, then $E:\mathrm{Int}\Omega^c\to\mathbb{R}$ ranges from $[d,c)$. Take $f:[d,c)\to\infty$ such that $F\circ E = \mathrm{Int}\Omega^c\to[0,\infty)$. Consider $(F\circ E)^{-1}(0)\cong\Omega^d$ is a topological manifold. There exists $\tilde{E}$ such that $\|\tilde{E}-F\circ E\|_{C^2}<\varepsilon$ and the critical points of $\tilde{E}$ is non-degenerate. Then the index of critical points of $\tilde{E}$ has index $\geqslant\lambda_0$.
Using Morse Theorem, $\mathrm{Int}\Omega^c(t_0,\cdots,t_n)$ is obtained by attaching cells to Ω^d, since the cells has dim $\geqslant\lambda_0$, $\pi_i(\mathrm{Int}\Omega^c(t_0,\cdots,t_n),\Omega^d)=0$ for small i.
Since $\mathrm{Int}\Omega^c(t_0,\cdots,t_n)$ is the deformation retract of $\mathrm{Int}\Omega^c$, and the colimit of $\mathrm{Int}\Omega^c$ is Ω, we have $\pi_i(\mathrm{Int}\Omega^c,\Omega^d)=0$.

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Complex Bott Periodicity

2026/6/5

Milnor *Morse Theory* Chapter23

Outline: First by fibration theory, we show that $\pi_i U(n)$ is stable, and that $\pi_i U(n) \cong \pi_i SU(n)$. Also by fibrations we connect $\pi_i U(2m)$ with $\pi_i G_m(\mathbb{C}^{2m})$. Finally, using Morse theory and by studying the structure of $U(n)$, we find the connection of $SU(n)$ and $G_m(\mathbb{C}^{2m})$.

Let $U(n)$ be the set of unitary matrices in $\mathbb{C}^{n\times n}$ with standard inner product, then $U(n)$ is a compact Lie group.

Lemma: The tangent space $TU(n)_I$ can be identified with the space of skew-Hermitian matrices.
Proof: Consider $\exp : \mathbb{C}^{n\times n} \to \mathbb{C}^{n\times n}, A \mapsto \sum_{n=0}^{\infty} \frac{1}{n!}A^n$, which is a local homeomorphism from $0 \in \mathfrak{u}(n) = \{A \in \mathbb{C}^{n\times n} : A + A^* = 0\}$ to $I \in U(n)$.
If $A+A^* = 0$, then take $\gamma : [0,1] \to U(n), t \mapsto \exp(tA)$, we have $\exp(tA) \in U(n)$ since $(\exp(tA))^* = \exp(tA^*) = \exp(-tA)$. Also $\gamma(0) = I$ and $\gamma'(0) = A$, so $A \in TU(n)_I$.
For any $A \in TU(n)_I$, there exists $\gamma : [0,1] \to U(n)$ such that $\gamma(0) = I$ and $\gamma'(0) = A$. Differentiating $\gamma(t)^*\gamma(t) = I$, we have $\gamma(t)'^*\gamma(t) + \gamma(t)^*\gamma'(t) = 0$, and at $t = 0$, $A^* + A = 0$. $\square$
Hence the Lie algebra $\mathfrak{g}$ of $U(n)$ can also be identified with skew-Hermitian matrices, since for any $A, B \in TU(n)_I$, $[A,B] = AB - BA$ corresponds to the bracket product of $\exp(tA)$ and $\exp(tB)$.

Note.
Construction of Riemannian Metric:
We explicitly construct a Riemannian metric on $U(n)$:
For matrices $A, B \in \mathfrak{g}$, define $\langle A, B\rangle = \operatorname{Re}\operatorname{tr}(AB^*)$, then it is an inner product, so it defines a unique left invariant Riemannian metric. To verify the metric is also right invariant, we check that it is invariant under $A \mapsto SAS^{-1}$.
The induced linear map is $\operatorname{Ad}(S) : TU(n)_I \to TU(n)_I$, given by $\operatorname{Ad}(S)A = SAS^{-1}$ since $\exp(SAS^{-1}) = S\exp(A)S^{-1}$. Hence the inner product is clearly invariant under $\operatorname{Ad}(S)$.

Observe that any $A \in \mathfrak{g}$ can be unitarily diagonalized as $TAT^{-1} = \operatorname{diag}(ia_1, \cdots, ia_n)$, and any $S \in U(n)$ can be unitarily diagonalized as $TST^{-1} = \operatorname{diag}(e^{ia_1}, \cdots, e^{ia_n})$. Hence $\exp : \mathfrak{g} \to U(n)$ is surjective.
Analogously, for $\mathfrak{g}' = \{A : A + A^* = 0, \operatorname{tr}(A) = 0\}$, the identity $e^{\operatorname{tr}(A)} = \det(\exp A)$ for $A \in \mathfrak{g}$ implies $\mathfrak{g}'$ is the Lie algebra for $SU(n)$, since $\det(\exp tA) = 1$ for all small t implies $\operatorname{tr}(A) = 0$.

Now we consider all geodesics of $U(n)$ from I to $-I$, i.e. all $A \in \mathfrak{g}$ such that $\exp(A) = -I$. Take $T \in U(n)$ such that TAT^{-1} is diagonal, then $\exp(TAT^{-1}) = T(-I)T^{-1} = -I$, so $\exp(A) = -1$ iff

the eigenvalues of A are in the form $\{k_1 i\pi, \cdots, k_n i\pi\}$ where k_i are odd integers.
The length of the geodesic $t \mapsto \exp(tA)$ is $|A| = \sqrt{\operatorname{tr}(AA^*)} = \pi\sqrt{\sum k_i^2}$, so A is a minimal geodesic iff $k_i \in \{\pm 1\}$. In this case, A is uniquely determined by $\operatorname{Ker}(A - i\pi I)$, which is an arbitrary subspace of $\mathbb{C}^n$. Hence the minimal geodesics from I to $-I$ correspond to subspaces of $\mathbb{C}^n$.
To make it simple, consider instead $SU(n)$ where $n = 2m$, then with the requirement $\operatorname{tr}(A) = 0$, $\operatorname{Ker}(A - i\pi I)$ must be an $m-$dimensional subspace of $\mathbb{C}^{2m}$, so the space of geodesics is homeomorphic to the Grassmann manifold $G_m(\mathbb{C}^{2m})$.

Tip.
Grassmann Manifold:
The Grassmann manifold $G_m(\mathbb{C}^{2m})$ consists of (W, V) such that $\mathbb{C}^{2m} = W \oplus V$ is an orthogonal direct sum, with the relative topology.
For each $(W_0, V_0) \in G_m(\mathbb{C}^{2m})$, consider all $m-$subspaces W such that $W \cap V_0 = 0$, then they correspond to linear maps $T : W_0 \to V_0$ where $W = \{w + T(w) : w \in W_0\}$. Hence this open set is homeomorphic to $\operatorname{Hom}_{\mathbb{C}}(W_0, V_0) \cong \mathbb{C}^{m^2}$, so we obtain a coordinate chart.
The transition charts are smooth (indeed holomorphic), so $G_m(\mathbb{C}^{2m})$ is a complex smooth manifold.

We will prove the following lemma:
Lemma: Any non-minimal geodesic in $\Omega(SU(2m); I, -I)$ has index at least $2m + 2$.

Since the $(n+1)-$skeleton X^{n+1} completely determines $\pi_n(X)$, by Morse Index Theorem we obtain (since $\Omega(SU(2m); I, -I)$ and $\Omega(SU(2m); I, I)$ are homotopy equivalent).
Theorem: $\pi_i(G_m(\mathbb{C}^{2m})) \cong \pi_i(\Omega(SU(2m))) \cong \pi_{i+1}(SU(2m))$ for all $i \leqslant 2m$.

Note.
Fibrations and long exact sequences:
Def: A **fibration** $p : E \to B$ is a map that satisfies the homotopy lifting property: for any homotopy $h : X \times [0,1] \to B$ and $\tilde{h}_0 : X \times \{0\} \to E$ lifting $h|_{X\times\{0\}}$, there exists (not necessarily unique) a homotopy $\tilde{h} : X \times [0,1] \to E$ lifting h such that $\tilde{h}|_{X\times\{0\}} = \tilde{h}_0$.
Given a fibration $p : E \to B$ and any $b_0 \in B$, the fiber is $F = p^{-1}(b_0)$.
Analogous to exact sequences of chain complexes, a fibration induces a long exact sequence of homotopy groups.
Theorem: A fibration $F \hookrightarrow E \xrightarrow{p} B$ induces a long exact sequence

$$\cdots \to \pi_i(F) \xrightarrow{\iota_*} \pi_i(E) \xrightarrow{p_*} \pi_i(B) \xrightarrow{\partial} \pi_{i-1}(F) \to \cdots$$

Proof: By homotopy lifting property, any fibration induces the exact sequence $\pi_i(F) \to \pi_i(E) \to \pi_i(B)$. Also by this property, there is a natural map from the path space $\Omega B \to F$ such that $\Omega B \to F \to E$ is homotopy equivalent to a fibration. Hence we obtain the Puppe sequence

$$\cdots \to \Omega^2 B \to \Omega F \to \Omega E \to \Omega B \to F \to E \to B$$

and by the natural isomorphism $\pi_n(X) \cong \pi_{n-1}(\Omega X)$, the required long exact sequence.

Using the theory above, we obtain the following isomorphisms:
For each m consider the fibration $p : U(m+1) \to \mathbb{S}^{2m+1}, (C_1, \cdots, C_{m+1}) \mapsto C_{m+1}$ with fiber $p^{-1}(e_{2m+1}) = U(m)$. It induces the long exact sequence

$$\cdots \to \pi_i U(m) \to \pi_i U_{m+1} \to \pi_i \mathbb{S}^{2m+1} \to \pi_{i-1} U(m) \to \cdots$$

since $\pi_i \mathbb{S}^{2m+1} = 0 \forall i \leqslant 2m$, for any $i \leqslant 2m-1$ we have $\pi_i U(m) \cong \pi_i U(m+1) \cong \pi_i U(m+2) \cong \cdots$ so we denote the stable homotopy group $\pi_i U = \pi_i U(m)$. Furthermore, $\pi_{2m} U(m) \to \pi_{2m} U(m+1)$ is surjective.

The fibration $p : U(m) \to \mathbb{S}^1, A \mapsto \det A$ with fiber $p^{-1}(1) = SU(m)$ gives the isomorphism $\pi_i U(m) \cong \pi_i SU(m)$ for $i \geqslant 2$.

By $\pi_i U(m) \cong \pi_i U(2m)$, the fibration $U(m) \to U(2m) \to U(2m)/U(m)$ (identify $U(m)$ as a subspace $A \mapsto \operatorname{diag}(A, I)$) gives the exact sequence $\pi_i U(m) \to \pi_i U(2m) \to \pi_i(U(2m)/U(m))$ hence $\pi_i(U(2m)/U(m)) = 0$.
Note that the Grassmann manifold $G_m(\mathbb{C}^{2m})$ can be identified as $U(2m)/(U(m) \times U(m))$, so the fibration $U(m) \to U(2m)/U(m) \to U(2m)/(U(m) \times U(m))$ gives isomorphisms $\pi_{i-1} U(m) \cong \pi_i G_m(\mathbb{C}^{2m})$.
Therefore,

$$\pi_{i-1} U = \pi_{i-1} U(m) \cong \pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m) \cong \pi_{i+1} U(2m) = \pi_{i+1} U.$$

Since $U(1) = \mathbb{S}^1$, $\pi_0 U = \pi_0 \mathbb{S}^1 = 0$ and $\pi_1 U = \pi_1 \mathbb{S}^1 \cong \mathbb{Z}$, therefore
Theorem (Bott): The stable homotopy groups $\pi_i U$ of $U(n)$ are periodic with period 2, and

$$\pi_{2n} U = 0, \quad \pi_{2n+1} U = \mathbb{Z}.$$

Proof of main lemma:
Lemma: Any non-minimal geodesic in $\Omega(SU(2m); I, -I)$ has index at least $2m+2$.
Proof: Recall the Lie algebra of $SU(n)$ ($n = 2m$) is $\mathfrak{g}'$ consisting of all skew-Hermitian matrices of trace zero, and a given $A \in \mathfrak{g}'$ corresponds to a minimal geodesic from I to $-I$ iff the eigenvalues are $i\pi k_1, \cdots, i\pi k_n$ where k_j are odd integers with sum zero.
Recall the conjugate points to I on the geodesic $t \mapsto \exp(tA)$ is given by the negative eigenvalues of $K_A : \mathfrak{g}' \to \mathfrak{g}', W \mapsto \mathcal{R}(A, W)A$.
Assume $A = \operatorname{diag}(i\pi k_1, \cdots, i\pi k_n)$ with $k_1 \geqslant \cdots \geqslant k_n$, then if $W = (w_{ij})$, we have

$$K_A(W) = -\frac{1}{4}[[A, W], A] = -\frac{\pi^2}{4}((k_i - k_j)^2 w_{ij})$$

Note that: for each $j < l$, the matrix E_{jl} with 1 at (j, l) and -1 at (l, j), is in $\mathfrak{g}'$ and an eigenvector of eigenvalue $-\frac{\pi^2}{4}(k_j - k_l)^2$; each E'_{jl} with i at (j, l) and (l, j) is in $\mathfrak{g}'$ and an eigenvector of eigenvalue $-\frac{\pi^2}{4}(k_j - k_l)^2$; each diagonal matrix of trace zero is an eigenvector of eigenvalue 0.
Hence the negative eigenvalues of K_A are $-\frac{\pi^2}{4}(k_j - k_l)^2$ with $k_j > k_l$, and the conjugate points of the geodesic $t \mapsto \exp(tA)$ are exactly

$$t = \frac{2}{k_j - k_l}, \frac{4}{k_j - k_l}, \frac{6}{k_j - k_l}, \cdots$$

which contributes $2\left(\frac{k_j - k_l}{2} - 1\right)$ to the index. Therefore

$$\lambda = \sum_{k_j > k_l} k_j - k_l - 2.$$

If γ is minimal then $k_j \in \{\pm 1\}$ so $\lambda = 0$ as expected. Otherwise for $n = 2m$
Case1: At least $m+1$ of k_j are negative, then at least one $k_j \geqslant 3$ so $\lambda \geqslant (m+1)(3+1-2) = 2m+2$.
Case2: There are m positive and m negative k_j, but not all are ± 1. Then at least one $k_j \geqslant 3$ and $k_l \leqslant -3$, so $\lambda \geqslant 2(m-1) + 2(m-1) + (3+3-2) = 2m+2$.
Therefore in both cases the index $\lambda \geqslant 2m+2$. $\square$

A note on Real Bott Periodicity:
The proof of stability and the Grassmannian carries over to $O(n)$ smoothly:
By the fibration $O(n-1) \to O(n) \to \mathbb{S}^n$ we have $\pi_i O(n-1) \cong \pi_i O(n)$.
By the fibration $SO(n) \to O(n) \to \mathbb{S}^0$ we have $\pi_i SO(n) \cong \pi_i O(n)$.
By $O(m) \to O(2m) \to O(2m)/O(m)$ we have $\pi_i(O(2m)/O(m)) = 0$.
By $O(m) \to O(2m)/O(m) \to G_m(\mathbb{R}^{2m})$ we have $\pi_i G_m(\mathbb{R}^{2m}) \cong \pi_{i-1} O(m)$.
Main failure: $\pi_i G_m(\mathbb{C}^{2m}) \cong \pi_{i+1} SU(2m)$ does not hold in the real case.
Here, the space of minimal geodesics is not $G_m(\mathbb{R}^{2m})$ but $SO(2m)/U(m)$.
Instead, Morse theory gives:

$$\Omega SO(2n) \simeq SO(2n)/U(n) \quad \Omega(SO(2n)/U(n)) \simeq U(n)/Sp(n/2) \tag{16.1}$$
$$\Omega(U(2n)/Sp(n)) \simeq BSp(n) \quad \Omega(BSp(n)) \simeq Sp(n) \tag{16.2}$$
$$\Omega Sp(n) \simeq Sp(n)/U(n) \quad \Omega\big(Sp(n)/U(n)\big) \simeq U(n)/O(n) \tag{16.3}$$
$$\Omega\big(U(n)/O(n)\big) \simeq BO(n) \quad \Omega BO(n) \simeq O(n). \tag{16.4}$$

Therefore

$$\pi_i O = \pi_{i+1} BO = \pi_{i+2} U/O = \pi_{i+3} Sp/U = \pi_{i+4} Sp \tag{16.5}$$
$$= \pi_{i+5} BSp = \pi_{i+6} U/Sp = \pi_{i+7} SO/U = \pi_{i+8} SO = \pi_{i+8} O. \tag{16.6}$$